Cost Effective Housing Techniques

Unit I: Housing Scenario, Finance and Planning

Syllabus

a) Housing scenario: status of urban housing, status of rural housing. b) Housing finance: existing finance system in India, government role as facilitator, status of rural housing finance, impediments in housing finance and related issues. c) Land use and physical planning for housing: planning of urban land, Urban Land Ceiling and Regulation Act, efficiency of building bye-laws, residential densities. d) Housing the urban poor: living conditions in slums, approaches and strategies for housing the urban poor.

1. Introduction to Housing

Housing is not just a shelter of four walls and a roof. It includes the **house, its surroundings, infrastructure services** (water, sanitation, power, roads) and **social facilities** (school, health, market) which together give a safe and healthy living environment. After food and clothing, shelter is the third basic need of man.

Importance of housing

- Gives **security, privacy and dignity** to a family; improves **health** and productivity
- Housing is a major part of the **national wealth** (about 20-30% of capital formation)
- The construction sector gives **employment** to millions, second only to agriculture
- Housing has strong **multiplier effect** on 250+ allied industries (cement, steel, bricks, paint, timber)
- Poor housing leads to slums, disease, crime and social unrest

Meaning of cost-effective housing

Cost-effective (low cost) housing means reducing the cost of construction by proper **planning, design, materials and techniques** without sacrificing **strength, durability and comfort**. It is NOT cheap or poor-quality construction.

Ways of reducing cost

* Use of **local materials** and substitutes for scarce materials
* **Improved techniques** (precast, prefabrication, filler slab, arches)
* **Efficient planning**: reduce built-up area, avoid wastage of space, optimum room sizes
* Reduce **wall thickness, use of rat-trap bond, hollow blocks**

* Standardisation, modular coordination and **mass production**
* Cost of land and services reduced through **planned layouts**, higher density

2. Housing Scenario in India

India has a huge **housing shortage** that is both **quantitative** (not enough dwelling units) and **qualitative** (unserviceable kutcha houses, congestion, no services).

Housing shortage (approximate figures)

Item	Figure / Remark
Urban housing shortage (Technical Group, 2012)	About **18.8 million** dwelling units (2012), 95% in EWS and LIG
Rural housing shortage (2012)	About **43.7 million** dwelling units
Slum population (Census 2011)	About **65 million** persons (17% of urban population), 13.9 million slum households
Households without proper latrine / water	Large share (improving under Swachh Bharat, Jal Jeevan)
Distribution of shortage	Mostly **EWS (Economically Weaker Section)** and **LIG (Low Income Group)**

- Figures change with every census / survey; for exams remember the **order of magnitude** and the trend: shortage is mostly among the poor
- Causes: **rapid population growth, urbanisation**, rural-urban migration, high cost of land and building materials, poor housing finance, poor planning and lack of services

3. Status of Urban Housing

- India's urban population has grown from about **28% (2001) to 31% (2011)** and is expected to cross 40% by 2030
- Cities have a **dual housing market**: a formal sector (private builders, public agencies) for rich and middle class, and an **informal sector** (slums, squatters, unauthorised colonies) for the poor
- **Shortage of land and its high cost** is the main problem in metros; affordable housing is pushed to the city edges
- **Slums:** about one in six urban Indians lives in a slum; **congestion** (more than 1 family per room) is high

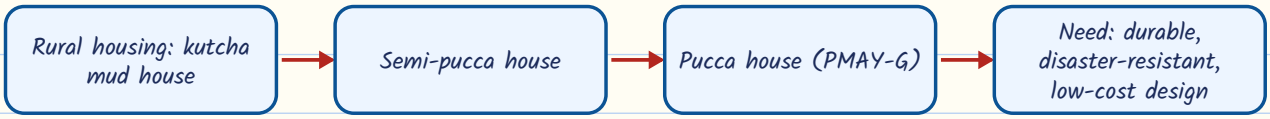
- **Rental housing** has declined because of rent control laws, which discouraged owners from building rental houses
- Government agencies: **State Housing Boards, Development Authorities, HUDCO, municipal corporations**; **PMAY-Urban (2015)** aims at housing for all

Housing categories by income (PMAY norms)

Category	Annual household income	Dwelling size (carpet area)
EWS	Up to Rs 3 lakh	Up to 30 sq.m
LIG	Rs 3 - 6 lakh	Up to 60 sq.m
MIG-I	Rs 6 - 12 lakh	Up to 160 sq.m
MIG-II	Rs 12 - 18 lakh	Up to 200 sq.m

4. Status of Rural Housing

- About **65-70% of the population** lives in villages; rural housing shortage is mainly **qualitative**: most houses are **kutcha or semi-pucca** (mud walls, thatch roofs)
- Types: **kutcha** (mud, bamboo, thatch), **semi-pucca** (brick walls, mud/tiled roofs), **pucca** (burnt brick / stone / concrete with RCC or tiled roof)
- Problems: poor **durability**, damage in floods and cyclones, **no sanitation, smoke from chulhas**, overcrowding, low income and no formal credit
- Rural houses are **self-built** with local labour and local materials; use of **local skills** is high but technology is poor
- Schemes: **Indira Awaas Yojana (IAY, 1985)**, now **PMAY-Gramin (2016)** with a target of 2.95 crore houses; unit assistance **Rs 1.2 lakh (plains), Rs 1.3 lakh (hills / difficult areas)**, house of minimum **25 sq.m** with a hygienic cooking space, support for toilet (SBM) and 90-95 days MGNREGA wage employment



5. Housing Finance

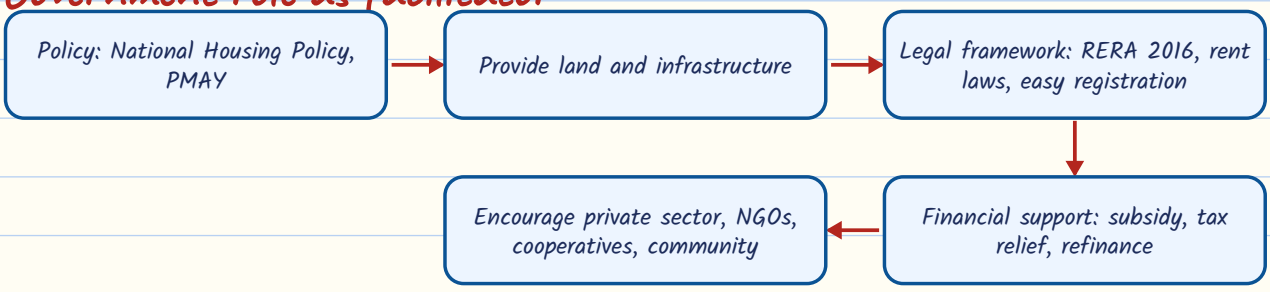
Housing is costly: a house may cost **5-10 times the annual income** of a family. Hence **long-term credit** is needed. **Housing finance** is the system that mobilises funds and lends them for purchase, construction or improvement of houses.

Existing finance system in India

Institution	Role
HUDCO (Housing and Urban Development Corporation, 1970)	Government techno-financial institution; loans to state boards and agencies, **EWS / LIG housing and slum improvement**, rural housing
National Housing Bank (NHB, 1988)	Apex regulator, subsidiary of RBI; refinances housing finance companies, regulates HFCs (now under RBI)
HDFC (1977), LIC Housing Finance, PNB Housing, Can Fin Homes etc.	**Housing Finance Companies (HFCs)** - private, market-based long-term loans
Commercial banks	Major share of housing loans; **priority-sector lending**
Cooperative banks, Housing cooperatives, NBFCs	Local lending, micro-finance
Employers, provident funds, GIC, LIC	Housing loans / advances to employees, funds for housing agencies
State housing boards and Development authorities	Build and allot houses with hire-purchase

- **Sources of funds:** deposits, bonds and debentures, **refinance from NHB**, government equity, **external agencies (World Bank, HUDCO overseas)**, **provident funds, insurance**
- **Government incentives:** income-tax deduction on loan principal and interest (Sections 80C, 24b), **Credit Linked Subsidy Scheme (CLSS)** under PMAY, **priority sector** status, **RBI norms on loan-to-value**

Government role as facilitator



- The government moved from a **direct builder** to an **enabler / facilitator**: it makes **land available, sets standards and bye-laws, gives subsidies and reforms laws**, while the people and the private sector build
- Instruments: **National Urban Housing and Habitat Policy**, **Real Estate (Regulation and Development) Act (RERA, 2016)** to protect buyers, **Affordable Housing Fund**,

interest subsidy, **Credit Risk Guarantee Fund**, **Rental Housing (ARHC)**, **land pooling**, FAR relaxation, single-window clearance

Status of rural housing finance

- Formal credit is **very limited** in villages: low and irregular incomes, no clear **land titles / mortgage**, no proof of income
- Sources: **NABARD** refinance, **Regional Rural Banks**, cooperative banks, **HUDCO** (rural housing loans), micro-finance institutions, **self-help groups (SHGs)**, and traditional money-lenders at high interest
- **Grants** under PMAY-G are the main support; loan up to **Rs 70,000** is permitted for the beneficiary under PMAY-G (subject to rules)
- Need: **micro-housing finance**, incremental lending, community guarantees, and land rights

Impediments in housing finance and related issues

- **Low incomes** of the target group; lack of **repayment capacity**; irregular, informal-sector earnings
- **Unclear land titles**, difficult mortgage and foreclosure laws; high **stamp duty and registration charges**
- High **interest rates**, high **loan-to-value** expectations, **long processing** and documentation, collateral needs
- **Shortage of long-term funds** (short-term deposits financing long-term loans: asset-liability mismatch)
- **Rent control laws**, building bye-laws not matching the poor, **delays in approval**, high cost of land
- **Non-performing assets**, project delays by developers, cost escalation
- Weak role of **cooperatives** and lack of **innovative products** for the informal sector

6. Land Use and Physical Planning for Housing

Land is the most scarce and costly input of housing. In big cities **land cost is 40-60%** of the cost of a house. Proper **land use planning** gives orderly development, efficient services and affordable housing.

Planning of urban land

- **Master Plan / Development Plan** (20 yrs) fixes **land use zones**: residential, commercial, industrial, recreation, public, transport, agriculture
- **Zoning regulations** control the use and density; **Zonal / Local area plans** (5 yrs), and **town planning schemes**
- **Neighbourhood concept (Clarence Perry):** a unit of about **5000 population**, with primary school at centre (walking 400-800 m), shops, park, and no through traffic
- Typical land distribution in a residential area: **residential 45-50%, roads 15-20%, open and recreation 15-20%, community facilities 10-15%**
- **Land policy tools:** acquisition, **land pooling and readjustment**, land banks, **Transfer of Development Rights (TDR)**, reservation of **EWS / LIG plots** (typically 15-25% of land / floor area in new projects), **sites and services** schemes, cross-subsidy
- **Strategies for land supply:** release of **government / surplus land**, redevelopment of old areas, **vertical growth with higher FSI**, optimum use of vacant land

Urban Land (Ceiling and Regulation) Act, 1976

- Passed to **prevent concentration of urban land** in few hands, to **prevent speculation and profiteering**, and to make **land available for housing the weaker sections**
- Fixed a **ceiling limit** on vacant land that a person could hold in an urban agglomeration (**500 to 2000 sq.m** depending on the city category); the **excess land** was to be taken over by the state and used for housing for the poor
- **Failures:** litigation, delays, **land hoarding and black market** instead of supply; **fall in land supply**; escape by legal loopholes; the Act did **not** produce housing for the poor
- **Repeal:** the **Urban Land (Ceiling and Regulation) Repeal Act, 1999** repealed the Act (adopted by states); land prices and supply improved afterwards in many places
- Present approach: **encourage land development by the private sector, regulation of land use**, not ceiling

Efficiency of building bye-laws

Building bye-laws are rules made by the local body to regulate construction for **safety, health, convenience** and orderly growth.

Provision	Meaning
FAR / FSI (Floor Area Ratio)	Total floor area / plot area; e.g., FAR 2.0 means built-up area twice the plot

Provision	Meaning
Ground coverage	Percentage of the plot covered by the building (30-60%)
Setbacks	Front, rear, side open spaces for light, ventilation and fire safety
Height of building	Limited by street width, airport rules, fire norms
Minimum room sizes	Habitable room 9.5 sq.m (NBC); kitchen 4.5 sq.m; bathroom 1.8 sq.m
Plot size	Minimum plot size for EWS reduced to 20-30 sq.m
Parking, staircase, lift, fire exits	Safety and services

- **Problems with existing bye-laws:** based on old standards, **too rigid and costly** for the poor, hinder **incremental housing**, high set-back / plot-size requirements, multiple clearances and delays
- **Reforms: model bye-laws, NBC 2016** compliance, **relaxed standards for EWS / LIG** (smaller plots and rooms, row housing, mixed use), **single-window, online approval**, higher **FSI near transit**, **performance-based** rules

Residential densities

Density = number of persons (or dwelling units) per hectare. It controls the **cost of land and services per family**.

Types of density

* **Net residential density** = population / area of residential plots only (excluding roads, parks)
* **Gross density** = population / total area including roads, open spaces, facilities
* **Neighbourhood density** and **city density**
* Dwelling units per hectare (**DU/ha**), **habitable rooms** per hectare, **FAR** / built-up area

Density type	Persons per hectare (approx.)
Low density	Less than 75 (bungalows)
Medium density	75 - 200
High density	200 - 500 (flats, row houses)
Very high (slums / congested)	More than 500

- **Higher density reduces the per-family cost** of land, roads, water, sewer lines, and electrification; typical Indian norms for planned cities are **100 - 300 persons / ha** (and plotted EWS schemes 300-500)
- Too high density: **congestion, lack of light and air, health, traffic, poor services**; so balance is needed

- **Plot size and density:** a 30 sq.m EWS plot gives about 250-300 DU/ha; 60 sq.m LIG plot about 100-125 DU/ha; multi-storey flats reach 300-600 persons/ha with open space left

7. Housing the Urban Poor

The urban poor (EWS and LIG) cannot afford formal housing: land, building cost and loan repayments are beyond their income. They live in **slums, squatter settlements, footpaths, rented rooms**.

What is a slum?

- As per the **Slum Areas (Improvement and Clearance) Act, 1956**, a slum is an area unfit for human habitation because of **dilapidation, overcrowding, faulty design, lack of ventilation, light, sanitation** or any combination of these
- Census 2011 definition: compact area of at least **300 population or 60-70 households** of poorly built, congested tenements in unhygienic conditions with inadequate infrastructure
- Types: **notified, recognised and identified slums**; squatter settlements; **pavement dwellers**; unauthorised colonies

Living conditions in slums

Aspect	Condition
Housing	Temporary / kutcha structures of plastic, tin, bamboo, mud; **overcrowded** (5-6 persons in a 10 sq.m room)
Water supply	Public taps / hand pumps, shared by many; irregular and unsafe
Sanitation	No toilets, **open defecation**, open drains, garbage dumped
Environment	Narrow lanes, no light and ventilation, flood prone low lands, near rail tracks, nalas
Health	High incidence of **TB, diarrhoea, malaria, dengue, skin diseases**; high infant mortality
Social / economic	Low and irregular income (informal work), **insecurity of tenure**, no credit, crime, fire hazards, poor education
Services	Few schools, clinics, no electricity or illegal connections

Approaches and strategies for housing the urban poor

Approach	Idea
Slum clearance and redevelopment	Demolish and rebuild (old approach); causes displacement and loss of livelihood, costly
In-situ slum upgradation	Improve **services and houses at the same place**: water, drains, toilets, lanes, security of tenure; minimal displacement
Slum rehabilitation / redevelopment with private participation	Land of slum used as resource; **free-sale component** cross-subsidises the free houses of slum dwellers (Mumbai SRA, **Slum Rehabilitation Scheme**)
Sites and services	Government provides **plots with water, sewer, roads** and the poor build their house **incrementally** with loans / own labour; cheap and flexible
Environmental Improvement of Urban Slums (EIUS)	Basic services, central sector (1972)
Relocation / resettlement	Shifted to the periphery; fails if far from jobs
Rental and dormitory housing	For migrants; **Affordable Rental Housing Complexes (ARHC)**
Community participation: NGOs, CBOs, SHGs, cooperatives	Beneficiary-led construction, savings groups

Government programmes (urban poor)

* **VAMBAY (2001)** Valmiki Ambedkar Awas Yojana; **NSDP**; **JNNURM – BSUP and IHSDP (2005)**
* **Rajiv Awas Yojana (RAY, 2011)**: slum-free India, **property rights** to slum dwellers
* **PMAY-Urban (2015)** four verticals: **In-situ Slum Redevelopment (ISSR)**, **Credit Linked Subsidy Scheme (CLSS)**, **Affordable Housing in Partnership (AHP)**, **Beneficiary-Led Construction (BLC)**
* **Basic Services to the Urban Poor (BSUP)**, Swachh Bharat, Smart Cities, **DAY-NULM** (livelihoods)

- **Strategies:** security of tenure **(pattas)**, incremental housing, convergence of schemes (water, sanitation, livelihood), participation, **low-cost technology and local materials**, planned land reservation (15-25%), **mixed income housing**, rental options, linkage to **employment**

8. National Housing Policies and Programmes

Policy / Programme	Key points
National Housing Policy 1988	First policy; target of **housing for all**, special focus on the poor, creation of **NHB**, shift to facilitator role
National Housing and Habitat Policy 1998	Public-private partnership, **cooperatives**, rural housing, use of **cost-effective technology**, role of **Building Centres**
National Urban Housing and Habitat Policy 2007	**Affordable housing for all**, 10-15% land or 20-25% FAR for EWS/LIG in new projects, **PPP**, special economic zones
Jawaharlal Nehru National Urban Renewal Mission (2005)	BSUP and IHSDP for the urban poor
PMAY (Urban and Gramin)	Housing for All by 2022 (extended)
Affordable Housing Fund (NHB, 2018)	Funding for the affordable housing, refinance of HFCs
RERA 2016	Registration of projects and agents, consumer protection

Building Centres

- Set up by **HUDCO** and **NGOs** from 1988 to **spread cost-effective technology** to the common man
- Functions: **production of building components** (blocks, tiles, doors), **training of artisans and masons**, **demonstration** of techniques, technical advice, **material banks**
- Examples: Nirmithi Kendras (Kerala), Building Centres, **CBRI (Roorkee)**, **SERC (Chennai)**, **BMTPC (Delhi)** - research and promotion of technology

9. Cost Components of a House and Planning for Economy

Component	Share of total cost (approx.)
Land	10 - 40% (depends on city)
Building materials	55 - 65% of construction cost
Labour	15 - 25%
Services (water, power, roads)	10 - 15%
Fees, taxes, finance charges	5 - 10%
Profit / overheads	5 - 10%

- **Walls 25-30%, roof and floors about 20%, foundations 10-15%, finishes 15-20%** - so savings in walls and roof give biggest gain
- **Planning for economy:** compact plans, **reduce circulation area**, avoid long passages, group the wet areas (kitchen-toilet) with common plumbing wall, use of **row housing / semi-detached** (common walls), standard **modular dimensions (multiples of 100 mm)**, simple elevations without projections, adequate **ventilation**
- Optimum **building height** (3-4 storeys walk-up without lift for EWS/LIG) - cuts cost of lifts and structure
- **Plinth area** to carpet area ratio about 0.8; plinth area rates are used for estimation

Layout types for low cost housing

Layout	Features
Row housing (terrace)	Houses share side walls, **less land and boundary**, cheap services; used for EWS/LIG; 15-20 houses per row max
Cluster housing	Houses around a common open court, **safe play and community space**, shorter road and service lengths
Detached / semi-detached	More land; costly
Walk-up flats (G+3 / G+4)	High density, shared staircase; **low cost of land per family**
Plotted development with sites and services	Plots 20-60 sq.m, incremental growth

Principles of site selection for housing

* **Accessibility** to work place and transport, **availability of services**
* Good **drainage**, no flooding, firm sub-soil, away from industrial pollution
* **Cost of land** and legal title clear; room for future expansion
* Near **schools, health and market**; avoid agricultural or forest land

10. Quick Revision Points

Remember for exam

* Housing shortage: urban about 18.8 million, rural about 43.7 million (2012); mostly EWS and LIG
* Government role has changed from builder to **facilitator / enabler**
* HUDCO 1970, NHB 1988, HDFC 1977; RERA 2016; PMAY-U 2015, PMAY-G 2016
* ULCRA 1976 repealed in 1999 because it failed to deliver land
* Density = persons per hectare; higher density = lower cost of services
* In-situ upgradation, sites and services and slum rehabilitation are the key approaches for the poor

Important questions

1. Explain the housing scenario in urban and rural India. What are the causes of housing shortage?
2. Describe the existing housing finance system in India and the role of government as a facilitator.
3. Discuss the impediments in housing finance and the status of rural housing finance.
4. Explain the planning of urban land. What are the objectives and drawbacks of the Urban Land Ceiling Act?
5. Discuss the efficiency of building bye-laws and suggest reforms for low cost housing.
6. Define residential density and explain its types and effects on cost.
7. Describe the living conditions in slums.
8. Explain the approaches and strategies for housing the urban poor.

Cost Effective Housing Techniques

Unit II: Low Cost Resilient Housing Technology

Syllabus

Development and adoption of low cost resilient housing technology: introduction - adoption of innovative cost effective construction techniques - adoption of precast elements in partial prefabrication - adopting total prefabrication of mass housing in India - general remarks on precast roofing / flooring systems - economical wall systems - single brick thick load bearing wall - 19 cm thick load bearing masonry walls - half brick thick load bearing wall - fly-ash gypsum brick for masonry - stone block masonry - adoption of precast RC plank and joist system for roof / floor in buildings.

1. Introduction

Housing cost can be cut by using **improved construction technology** that saves **materials, time and labour**, and also gives a **resilient** (disaster-resistant, durable) house. Development of such technology in India has been led by **CBRI (Roorkee), SERC (Chennai), NBO, BMTPC, HUDCO Building Centres, IITs** and NGOs.

Principles of cost reduction in technology

- **Save materials**: reduce thickness of walls and slabs, use of **hollow, perforated, cored elements**
- **Replace costly materials** (steel, cement, timber) by local and waste based materials (lime, fly ash, bamboo, stone, soil)
- **Speed**: factory made precast parts reduce construction period and interest on money
- **Standardisation and modular coordination** (100 mm module), repetition of the same unit gives mass production
- **Skill and tools**: simple methods that local masons can learn; **less wastage**
- **Appropriate technology**: suited to local climate, materials, culture, and disasters (earthquake, flood, cyclone)

Resilient housing means

* Strong against **earthquake, wind, flood, fire** - safe life and property
* **Durable** with low maintenance; **thermally comfortable**

* Quick **repair and recovery** after the disaster
* **Sustainable**: low embodied energy and carbon

2. Innovative Cost Effective Construction Techniques

A. Foundation techniques

Technique	Details and saving
Arch foundation	Brick arches between footings carry load of wall; saves 30-40% of masonry and concrete in foundation
Brick (stone) on edge foundation	Bricks laid on edge for light loads; saves 15-20%
Under-reamed pile foundation	Bored pile with 1-2 **bulbs (under-reams)** at base; for **black cotton / expansive soil** and loose soil; pile dia 200-300 mm, length 3.5 m or more; bulb dia 2.5-3 times shaft; **bulbs prevent uplift**
Raft / strip footing	For weak soils and with plinth beam
Rubble trench foundation	Trench filled with stone and lime; good drainage, low cost
Plinth reduced in height of 30-45 cm	Saves filling

B. Walling techniques

- **Rat-trap bond** - bricks laid on edge with a **cavity** in the wall (like rat trap); saves **25% bricks and 10-15% cost**, good thermal insulation, 230 mm wall (same strength); needs mortar 20-25% less
- **Cavity wall, hollow concrete / clay block, solid blocks (stabilised soil blocks, fly ash bricks, AAC blocks)** - big size gives fast work
- **Thin walls: half-brick and 19 cm** walls (see below), and **stone masonry in lime / cement-sand** with bonding courses
- **Ferrocement / precast wall panels, cellular lightweight concrete (CLC)**

C. Roofing and flooring techniques

Technique	Details and saving
Filler slab	Slab with **filler materials (clay pots, tiles, bricks, hollow blocks)** in the tension zone (concrete there is not needed); saves **10–25% cost** of concrete and steel, and improves insulation
Brick arch / Jack arch roof	Shallow brick arches between steel / RCC joists; no shuttering; saves 20–30%
Funicular shell	Shell of funicular (catenary) shape in compression; saves 30–40% concrete and steel
Ferrocement channel / shell units	Thin, light channels spanning 3–4 m, laid side-by-side, finished with screed; saves 30–35% cost vs RCC slab
Precast RC plank and joist	Described later (very popular)
Cored / hollow-core slab	Voids in precast slab reduce weight
Madras terrace, tile roofing on rafters, GI / AC sheet roof	Traditional / lightweight systems
Waffle slab, ribbed slab	Material saved in non-structural zones

D. Other economies

- **Doors and windows: RCC / ferrocement frames and shutters**, steel sections, **frameless doors**, or **plain wooden planks from plantation timber**; use sliding / pivoted shutters; **avoid lintel by arches**; **RCC lintel cum sunshade**
- **Finishing: no plaster** (rat-trap with pointing), **lime-cement mix plaster, cement-wash**; stabilised mud plaster; **neeru finish**; **IPS flooring**, brick floors, **mosaic tiles** only where needed
- **Services: combined plumbing wall**, **PVC pipes**, rain-water harvesting, solar water heaters, **CFL / LED** lighting

3. Adoption of Precast Elements in Partial Prefabrication

Prefabrication is the manufacture of building parts in a **factory or casting yard**, which are then transported and **assembled at site**. In **partial prefabrication** only some elements (lintels, window sills, door frames, planks, joists, stairs, sunshades, beams, columns, chajjas) are precast, and the rest (walls, foundations) are built in the usual way.

Precast element	Notes
Lintels, sunshades, chajjas, sills	Common, cast in mould, cut site work
RC door and window frames	Saves timber; durable; termite proof
Precast RCC planks and joists	For roof / floor (see section 7)
Stair flights, landings	Quick fixing
Precast columns, beams	Frames, moment connections
Ferrocement shells, channels	Roofing
Cover blocks, kerbs, manhole covers, septic tank rings	Small precast products
Hollow concrete and solid blocks, paving blocks	Block making machines

Advantages and limits of partial prefabrication

* **Saves** shuttering, centering (up to 80%), time, and **cement and steel** by quality control
* Less site labour and skill; **better quality** (cured in yard); **fast** construction
* Components can be **stored and reused**; construction continues in rains
* **Limits:** transport and **crane need**, joints (weak points), initial investment for moulds, and demand must be big enough

4. Total Prefabrication of Mass Housing in India

In **total (full) prefabrication** the whole building (walls, floors, roof, stairs) is made from factory-produced **large panels or boxes**, assembled by cranes and joined at site. It is useful for **mass housing** (hundreds of identical units), because the high cost of the plant is spread across many units.

Systems of prefabrication

System	Description
Large panel system	Storey-high **wall panels and floor slabs** cast in a plant, erected by crane; dry or wet joints
Frame (skeleton) system	Precast **columns and beams** with floors and non-load-bearing panels
Box (modular) system	Complete room units (boxes) made, stacked at site
Slab-column (lift slab) system	Slabs cast at ground and lifted
Mixed / hybrid system	Cast-in-situ core with precast floors and walls

Experience in India

- **Early attempts (1960s – 1980s):** HUDCO and NBO, **Hyderabad, Delhi, Mumbai, Pune** large panel projects; **Hindustan Prefab Ltd (HPL, Delhi)**; **Bombay Mhada**; many were stopped due to **high cost, transport problems, and lack of demand continuity**
- **Modern: Tunnel form (Mivan) / aluminium formwork** - monolithic casting of walls and slabs; **precast light weight panels (EPS cement sandwich panels, GFRG)**; **Light Gauge Steel Framed (LGSF) structure**; **Light House Projects (LHP) under GHTC-India 2019-21**: Indore (precast sandwich panel), Rajkot (monolithic concrete with tunnel forming), Chennai (precast concrete system), Ranchi (3D volumetric precast), Agartala (steel frame with GFRG panels), Lucknow (PVC panel + steel frame)
- **Prefab policy:** BMTPC promotes **Performance Appraisal Certification Scheme (PACS)** for new materials and technologies, **Technology Sub Mission under PMAY(U)**

Requirements for success of total prefab

* **Large scale and continuous demand** (at least 500-1000 units / year per factory)
* **Standardisation, modular design** and good planning of **joints** (earthquake and weather proof)
* **Transport and cranes; plant within 30-40 km** of the site; skilled team and quality control
* Government support: **bulk orders**, finance, and approval of codes

Merits and demerits of total prefabrication

Merits	Demerits
Fast (40-50% time saving), **less labour**, **superior quality**	High **initial cost** of plant, moulds and cranes
Less wastage, **less water** at site, clean site	**Transport** cost of heavy elements; roads
Repetition gives economy for large projects	Monotony in design, less flexibility; **difficult joints**
All-weather work; supervision easy	Needs skilled engineers; **uneconomical for small jobs**

5. General Remarks on Precast Roofing and Flooring Systems

- Roof and floor costs about **20-25% of the building cost**, so precast systems are widely used
- **Requirements:** structural strength to carry load and **diaphragm action** (earthquake forces must be transferred to walls), **light weight**, easy handling by **2-4 persons**, no shuttering and propping (or only light props), **thermal insulation, fire resistance**, good appearance of the ceiling, and **proper bearing** on walls (min **90-100 mm**) with a **bond beam**
- **Types:** (1) **RC plank and joist**, (2) **channel units** (cored, ferrocement), (3) **waffle / ribbed panels**, (4) **hollow-core planks and double-T slabs**, (5) **precast RC cored units (Cellular)**, (6) **L-panel, Jack arch with joists, brick panels (reinforced brick panel)**, (7) **Precast hollow clay tile slabs** (Madras terrace type)
- **Joining:** use **cast-in-situ screed (35-50 mm) over the units with mesh**, grouting of joints, **shear keys**, and tying with the **RCC bond beam** at roof level, for earthquake safety

6. Economical Wall Systems

Walls take 25-35% of the cost of a house. So economy of walls is the first target. Wall thickness is reduced where only **light loads** (single / double storey) are carried.

A. Single brick thick load bearing wall (230 mm)

- The **standard wall** of 230 mm (9 inch) in **English bond** in cement mortar **1:6**; used for **load bearing** up to **3-4 storeys**
- Economy by using **rat-trap bond** (saves 25% bricks), mortar 1:6 or 1:8, **no plaster**, or **lime-cement** mortar
- Strength: brick of 75 kg/cm^2 (7.5 N/mm^2) class gives permissible stress about **0.5 N/mm^2 (IS 1905)**
- **Slenderness ratio** limits: effective height / thickness not more than **27** (for 230 mm wall = about 3.1 m height)

B. 19 cm thick load bearing masonry wall

- A wall of **19 cm** thickness is made from the **brick of nominal size 19 x 9 x 9 cm** (modular) laid with **brick on header** (one brick = 19 cm), with **1:6** mortar
- Used for **single-storey and two-storey** buildings; **saves 17% of brickwork** vs 230 mm (so 15-20% on wall cost), and reduces dead load, increases floor area
- Strength is enough for **low rise (up to 2 storeys)**: slenderness ratio checks; must be used with **RCC bands** at plinth, lintel and roof level, and **vertical reinforcement at corners** in earthquake zones
- Developed by **CBRI Roorkee**; for Zone III and below; **mortar 1:6** to 1:4; height of wall not more than **3.0 m**

C. Half brick thick load bearing wall (115 mm)

- Wall of **half brick** (stretcher bond) 11.5 cm thick: used as **partition walls** and also load bearing in **single storey buildings** in light roofing, with **short span (3.0 - 3.5 m)**
- Needs **stiffening** by **buttresses / piers** at 1.2 m, **cross walls**, vertical bars, or **RCC / brick-on-edge bands**; panel size limited; **height up to 2.5-3 m; length limit 3.5 m between cross walls**; mortar 1:3 to 1:4
- Saves **50% of bricks and mortar** as compared to 230 mm wall; reduces **dead load (good for seismic)**; but poor in thermal insulation and sound, and **weak in lateral load**, so used with **reinforced brickwork** (bars in every 4th course, **Brick-bat bands**) in seismic area
- Where loading is lighter, **cavity walls** may use 115 + 115 mm leaves with 50 mm gap

Wall type	Thickness	Use	Saving
Standard brick wall	230 mm	Multi-storey load bearing	—
Rat-trap bond	230 mm (cavity 75 mm)	Up to G+2	25% brick, 10-15% cost
19 cm wall	190 mm	Up to 2 storeys	15-20% of walling cost
Half brick wall	115 mm	Partition, single storey	Up to 50% bricks
Fly ash / AAC blocks	100-200 mm	Infill / load bearing	Light, thermal comfort

D. Fly ash - gypsum (FAL-G) bricks for masonry

Fly ash (a waste from thermal power stations) is mixed with **lime and gypsum**, producing **FaL-G bricks**. A pozzolanic reaction forms **calcium silicate and sulphoaluminate** hydrates that give strength.

FaL-G brick facts

* Mix: **fly ash 50-60%, lime 10-20% (or cement), gypsum 5-10%, stone dust / sand 20-30%**; moulded under **pressure**, **water cured** (not burnt, saves coal and top soil)
* **Size 230 x 110 x 75 mm (or modular)**, **compressive strength 7.5 - 12 MPa (IS 12894)**, water absorption below 20%
* **Light weight** (1.4-1.6 g/cc), smooth, accurate; **plaster can be thin or omitted**; **10-20% cheaper** than burnt clay bricks
* **Environment:** uses an industrial waste; **no top soil** is lost; less CO_2; **Fly ash brick use mandatory** (MoEF notification) within 300 km of thermal plants

- **Fly ash blocks (hollow / solid) and fly ash gypsum thick walls** are used for load bearing and infill walls: the **thick (230 mm) wall** from large FaL-G bricks provides **good thermal insulation**
- **Precautions:** adequate curing (7-14 days), avoid dry shrinkage, use cement-lime mortar, **do not use in contact with water** without damp-proof course

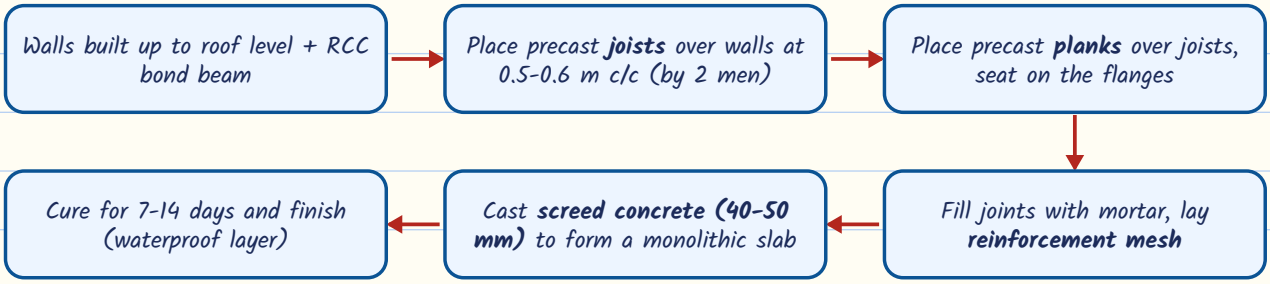
E. Stone block masonry

- Where **stone** is available (hilly and plateau regions), **dressed stone blocks** (large regular blocks, say **300-450 mm long, 200-250 mm high, 200-300 mm thick**) are used instead of random rubble; **quick laying, less mortar (saves 30-40% mortar)** and fewer joints; **walls may be 300-400 mm** thick; **stone is available free** at local quarries

- **Types:** ashlar masonry, **squared rubble**, **block-in-course**; **cut stone blocks of 3 cm face**; **laid with cement-sand 1:6** or lime mortar, with **through stones** at every 1.2 m, **corner stones** and **horizontal bands**
- **Stone block** can be **hollow stone (cut out voids)**, or **concrete blocks made of stone chips**; with **bonding and RCC bands**, they work well in earthquake and give a **cool house**
- **Economy:** local stone, less mortar, strong, **durable**; **disadvantage:** heavy, high transport, low thermal insulation if thin; needs skilled stone cutters

7. Precast RC Plank and Joist System for Roof / Floor

A **precast RC plank and joist** system was developed by **SERC (Chennai) and CBRI** for **low-cost roofing** of buildings with spans up to **3–4 m**. The roof has **precast joists (small beams)** laid at about 500–600 mm centres, with **precast planks** (thin slabs) resting on joists; then a **cast-in-situ screed (30–50 mm)** with nominal reinforcement.



Item	Typical details
Planks	Size **300 x 500 x 40–50 mm**; M20 concrete, **2 bars of 4–6 mm**; weight 10–30 kg: carried by one man
Joists	Size **100–130 x 150–200 mm**; **T or L section** with stirrups, reinforcement 2 bars of 8–10 mm; length up to 3.5 m
Screed	**1:2:4 concrete or M20**, 40–50 mm thick with **weld mesh / 6 mm bars @ 200 mm**
Span	Up to **3.6 m**
Saving	**25–30% on cost** and 30–35% in steel and cement vs. RCC slab; no shuttering, only light props for joists (removed after 14 days)
Variants	Joist + **hollow clay tile / pot / filler block** system; **planks of ferrocement, brick, or lightweight**; joist + stone slab (precast)

Merits and precautions

* **Merits:** saves shuttering and centering, quick, low cost, works by local labour, **good quality** of precast elements, easy repair
* **Precautions:** proper **bearing (at least 100 mm)**, joists should be **propped** until the screed gains strength, **joints** filled well, **tie the roof with bond beam** for earthquake, **waterproofing** treatment, edge reinforcement

8. Cost Saving: A Worked Comparison

Example: comparison of RCC slab vs. plank and joist roof for 3 m x 4 m room

RCC slab (100 mm): concrete = 12 x 0.1 = 1.2 m^3; steel about 60 kg (50 kg/m^3 x 1.2) + shuttering (12 sq.m).

Plank & joist: planks 0.8 m^3 + screed 0.5 m^3 + joists - cuts concrete by 30% and steel by 40%; no shuttering (props only) => cost saving of about **25-30%** and time saving of 2-3 weeks.

9. Modular Coordination and Standardisation

- **Modular coordination** = dimensioning of building and its components in multiples of a basic **module M = 100 mm** (IS 1233), so that parts from different makers fit together without cutting
- Preferred multi-modules: **3M (300 mm), 6M, 12M**; **brick size 190 x 90 x 90 mm** (modular) vs 230 x 110 x 75 mm (traditional)
- **Benefits:** less wastage, reduces **site cutting**, free interchange, **mass production** of doors, windows, planks, **easier design and estimating**
- **Standard components:** door 1000 x 2100 mm, window 1000 x 1200 mm, **ventilators 600 x 450 mm**, stair width 0.9-1.0 m, rise 150-170 mm, tread 250-275 mm

10. Resilience: Safety Features in Low Cost Technology

Feature	Why / how
RCC / reinforced brick bands	Plinth, lintel, roof (and sill) bands tie walls like a **box** against earthquake (IS 4326)
Vertical steel at corners and jambs	Prevents separation of walls at corners
Regular, symmetrical plan, light roof	Less seismic force, no torsion
Wall length / opening limits	Openings not more than 1/3 of wall length, 600 mm away from corners
Cyclone / wind	Hip roof with slope 30 degrees, **tie-down straps**, strong foundations, low height
Flood	Raised **plinth**, stilts, water resistant materials, stabilised blocks
Fire	Non-combustible materials, fire-retardant treatment of thatch
Quality precast joints	Wet joints, shear keys, **dowels and bolts** transfer the diaphragm forces

- **Precast units must be tested** for load (transverse strength) and dimensions; **cube strength** of concrete (M20 or more), **curing 14 days** (water curing), **stacking on level ground**, lifting at the marked points
- **Cost control** by **bulk purchase, sharing mould sets (use 50-100 times), semi-skilled labour, and local production through Building Centres**

11. Quick Revision Points

Remember for exam

* Rat-trap bond saves about 25% bricks; filler slab saves 10-25%; plank and joist saves 25-30% (no shuttering)
* Under-reamed piles are used in expansive soils; arch foundation saves 30-40%
* 19 cm wall saves about 17% of brickwork; half-brick wall needs stiffening and bands
* FaL-G brick = fly ash + lime + gypsum (+ sand), cured not burnt, 7.5-12 MPa
* Total prefabrication needs mass demand: large panel, frame, box systems; Light House Projects (2021)
* Modular coordination (100 mm module), standardisation and bond beams are the essentials of safe precast

Important questions

1. Explain the innovative cost-effective construction techniques for foundation, walls and roofs.
2. Describe partial prefabrication with precast elements and its advantages.
3. Discuss the total prefabrication of mass housing in India with the types of systems.
4. Write short notes on precast roofing and flooring systems.
5. Explain economical wall systems: 230 mm wall, 19 cm wall and half-brick wall.
6. What are fly-ash gypsum bricks? Explain their composition and uses.
7. Explain stone block masonry and its advantages for cost-effective housing.
8. Describe the construction of the precast RC plank and joist roof / floor system with a neat sketch.

Cost Effective Housing Techniques

Unit III: Alternative Building Materials and Low Cost Infrastructure

Syllabus

Alternative building materials for low cost housing: introduction - substitute for scarce materials - ferro-cement - gypsum boards - timber substitutions - industrial wastes - agricultural wastes - alternative building maintenance. Low cost infrastructure services: introduction - present status - technological options - low cost sanitation - domestic waste - water supply - energy.

1. Introduction

Conventional materials like **cement, steel, burnt bricks, timber and sand** are becoming **scarce and costly** and their production consumes much **energy** and damages the environment (top soil loss, forest cutting, CO_2). So **alternative building materials** made from **local, renewable, waste** resources are needed to cut the cost of housing and protect the environment.

Desirable properties of an alternative material

* **Cheap** and **locally available**, easy to produce by simple tools and local skill
* Adequate **strength, durability, fire and weather resistance**
* **Low embodied energy** (energy used in making) and eco-friendly
* **Acceptable** to the people; standardised; Indian Standards (BIS) or **BMTPC PACS** appraisal

Material	Embodied energy (MJ/kg, approx.)
Steel	25 - 35
Aluminium	150-200
Cement	4 - 5
Burnt clay brick	2 - 3
Concrete (plain)	1 - 1.5
Fly ash brick	0.5 - 1
Stabilised soil block	0.7 - 1
Timber (local)	0.5 - 2
Bamboo, mud, straw	Very low (0.1 - 0.5)

2. Substitutes for Scarce Materials

Scarce material	Substitutes
Cement	**Lime-pozzolana (lime + fly ash / surkhi)**, **Portland pozzolana cement (PPC)** with fly ash, **slag cement (PSC)**, rice husk ash cement, **magnesium oxychloride (Sorel) cement**, **lime-fly ash-gypsum**, clay-based mortars, **alkali-activated (geopolymer)** binders
Steel	**Bamboo reinforcement**, **ferrocement** (wire mesh), **prestressed concrete**, **HYSD / TMT bars (less quantity)**, **brick reinforcement**, GFRP / basalt bars, plastic mesh
Timber	**Bamboo, RCC / ferrocement, steel, aluminium, FRP, PVC, particle board**, see section 5
Burnt clay bricks	**Fly ash bricks, FaL-G, sand-lime bricks, stabilised soil blocks, concrete blocks, AAC blocks, stone**, slag bricks
River sand	**Manufactured sand (M-sand)** from crushed stone, **quarry dust, fly ash, crushed recycled aggregate**
Coarse aggregate	**Slag, clinker, recycled C&D; waste, light weight aggregate (fly-ash aggregate, expanded clay)**
Roofing sheets / tiles	**Ferrocement / micro-concrete tiles (MCR), fibre-cement sheets, bamboo mat corrugated sheets, thatch** with treatment, recycled plastic
Plaster	**Gypsum plaster, mud-lime plaster, cement-fly ash mortar**

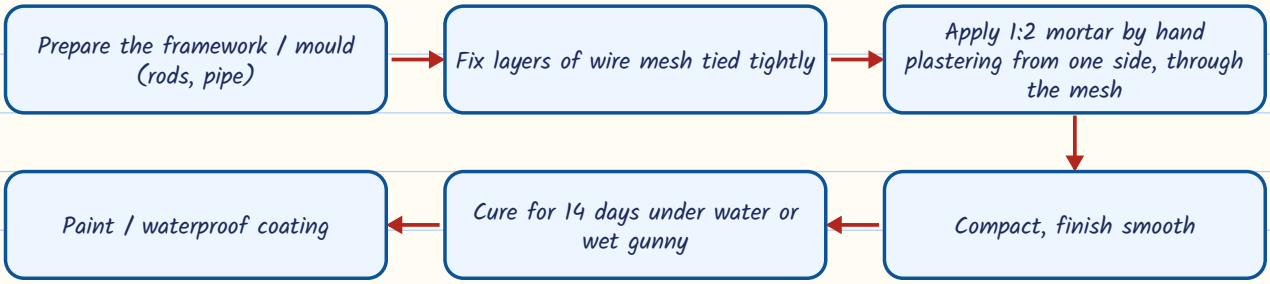
Lime and lime-pozzolana

- **Lime** (from limestone / kankar) is an old binder; **hydraulic lime** sets under water; **fat lime + pozzolana** (fly ash, surkhi (burnt-brick powder), rice husk ash, calcined clay) gives strength slowly, **cheaper than cement and consumes less energy**
- **Lime-pozzolana mortar 1:2** for masonry and plaster, **lime concrete** for foundations and roof terracing (**lime terracing** with brick-bats, jaggery, fenugreek, **Madras terrace**) - traditional, durable, breathable
- Also good for **repair of old buildings and heritage** structures

3. Ferro-Cement

Ferrocement is a thin composite of **rich cement-sand mortar** reinforced with **closely spaced, multiple layers of fine wire mesh** (and sometimes small-diameter rods). The thickness is only **10 – 40 mm**; and it behaves like homogenous elastic material with very high **crack control and impact resistance**.

Component	Details
Mortar	Cement : sand = **1 : 1.5 to 1 : 2.5**; **w/c 0.4 - 0.45**; sand passing 2.36 mm, fine; with **plasticiser**; strength M30 - M40
Wire mesh	**Welded mesh or woven (chicken) mesh, hexagonal**, gauge 20-24, opening 6-25 mm, galvanised; **2-6 layers**
Skeletal steel	Rods 4-6 mm @ 50-100 mm to form framework
Volume fraction of steel	**2 - 8%** (about 300 - 500 kg per m^3)
Thickness	**10 - 40 mm** (covers 2-3 mm)
Curing	Water cure **7 - 14 days** (very important, thin section)



- **Applications in housing: roofing elements (channels, folded plates, shells and domes), wall panels and partitions, water tanks (storage), grain bins (silos), septic tanks, manholes, sunshades, door and window frames, shutters, lintels, latrine units, bio-gas holders, boats, furniture, pipes, precast ferrocement roofing units - "channel roof"**
- **Advantages:** very **thin and light** (low dead load), **cheap** with local skill, **no formwork** (or little), tough and **crack resistant**, easy to repair, **fire and rot proof**, **can take any shape**, 30-50% cheaper than RCC for thin members
- **Limitations: labour intensive**, **corrosion** of the mesh (needs good cover and treatment), difficult to **join and fasten**, **local buckling**, poor fire resistance (thermal conductivity), skilled work needed, no codes in many cases, **heat gain** in the roof

4. Gypsum Boards

Gypsum ($CaSO_4.2H_2O$) is a mineral found in Rajasthan, J&K;, Tamil Nadu; and **by-product gypsum (phospho-gypsum, FGD gypsum)** from industries. When heated at about **150 degrees C** it becomes **plaster of Paris (hemi-hydrate)** that sets when mixed with water.

Gypsum reaction

$CaSO_4.2H_2O$ --(heat 150 C)--> $CaSO_4.0.5H_2O + 1.5\ H_2O$ (calcination)

$CaSO_4.0.5H_2O + 1.5\ H_2O$ -> $CaSO_4.2H_2O$ (setting, in 8 to 15 minutes)

Gypsum product	Description / use
Gypsum plaster board (paper-faced)	Core of gypsum between two **paper liners**, thickness **9.5, 12.5, 15 mm**, size 1200 x 2400 mm; **ceilings and partitions** (dry wall); fixed on **GI metal studs**
Fibrous gypsum board / glass fibre reinforced gypsum (GFRG - "Rapidwall")	Panels with glass fibre in gypsum, **core filled with concrete and reinforcement** - load bearing walls and floors in mass housing
Gypsum plaster	Single coat **internal plaster 10 mm**, smooth, no curing, **saves 40-50% time and 20-30% cost** over cement plaster
Gypsum blocks and tiles	Light partition blocks, **false ceiling tiles**
Moisture resistant and fire resistant boards	Special additives for bathrooms and fire ratings (1-2 hours)
Cornices, mouldings	Decorative

- **Merits: light** (25 kg/m^2 for partition), **fast dry construction**, good finish, **fire resistant**, **thermal and acoustic insulation**, no curing water, uses **waste gypsum**; reduces dead load (good for earthquake)
- **Demerits: not for wet areas without treatment** (loses strength in water), low impact resistance, **non-structural** (boards), the metal frame needs protection; **skilled fixing**, imported systems costly

5. Timber Substitutions

Timber is costly and scarce because of **deforestation**. Doors, windows, frames, roof trusses and furniture can use alternative materials.

Substitute	Notes
Bamboo	Fast-growing, **strength comparable with steel by weight**; treated by **borax-boric acid** or smoke; used for **roof trusses, walls (ekra), scaffolds, bamboo reinforcement, flooring**; Bamboo mat boards and **bamboo mat corrugated sheets (BMCS)**
Plantation timber and treated softwood	Eucalyptus, rubber wood, poplar, **preservative treated** (CCA, boron)
Particle board, MDF, plywood (plantation), block board	Made from wood chips and **agri-residues** with resin

Substitute	Notes
RCC / precast concrete door and window frames, lintels	Durable, termite proof; no polishing
Steel and aluminium sections / frames	Rolled sections, pressed steel frames; **PVC and FRP**
Ferrocement / fibre-cement doors	Thin shutters, cheap
FRP (GRP), PVC / WPC (wood plastic composites)	Rot and termite proof, light
Jute-polymer composites, coir boards, wood-wool cement boards	Panels for partition, ceiling

- **Design economy:** use of **smaller sections**, **built-up sections, composite trusses**, **doors without frame (the frameless door)**, **sliding windows**, reduce area of shutters, planning for **modular door and window** sizes; **use RCC lintel-cum-sunshade**

6. Industrial Wastes in Building Materials

Industry produces millions of tonnes of **solid wastes** which pollute land, water and air. Using them in construction **saves cost and natural resources**.

Waste	Source	Use in construction
Fly ash	Thermal power plant (above 200 million t/yr in India)	**PPC cement, fly ash bricks, blocks, concrete admixture, lime-pozzolana, road embankments, light weight aggregate, FaL-G, cellular concrete**
Blast furnace slag (GGBS)	Steel plant	**Slag cement (PSC), slag bricks, aggregate,** geopolymer
Steel slag	Steel plant	Road base, aggregate
Phospho-gypsum, FGD gypsum	Fertiliser / power plants	**Gypsum plaster, blocks, boards, cement retarder**
Red mud	Aluminium (bauxite) industry	**Bricks, tiles, red mud-polymer composite doors / windows (BMTPC),** cement
Silica fume	Silicon metal industry	High performance concrete
Mine tailings, marble slurry, stone dust	Mining and stone cutting	**Bricks, blocks, M-sand, fillers**
Construction and demolition (C&D;) waste	Demolished buildings	**Recycled aggregate, bricks, paver blocks, non-structural concrete**

Waste	Source	Use in construction
Waste plastics, tyre rubber	Domestic / industrial	Plastic roads, tiles, rubberised concrete, lightweight panels
Spent foundry sand, ceramic waste, glass cullet, bagasse ash	Industry	Blocks, tiles, aggregates

Benefits of using industrial wastes

* Cuts cost and consumption of **cement, sand, clay** (conserves top soil and energy)
* **Solves disposal problem**, reduces pollution and landfill needs
* Many wastes improve the **durability** (low heat of hydration, resist sulphates)
* **Cautions:** quality control, **leaching of heavy metals**, radioactivity checks; code compliance

7. Agricultural Wastes in Building Materials

India produces **about 500 million tonnes of agricultural residue** each year; much is burnt (air pollution). They are **renewable and cheap**, and have low embodied energy.

Waste	Products
Rice husk / rice husk ash (RHA)	RHA has **85–90% amorphous silica** – a pozzolana: **lime–RHA cement, RHA concrete, bricks, blocks, boards, insulation**
Sugarcane bagasse and bagasse ash	**Particle boards, bagasse-cement boards, ash in concrete, fibre**
Wheat / paddy straw	**Straw bales** (walling), **straw boards (Stramit)**, mud-straw plaster, **thatch roof**
Coconut husk and coir	**Coir fibre boards, coir-cement, mats, ceilings**
Groundnut shell, cotton stalk, jute stick, sawdust, wood waste	**Particle boards, wood-wool cement boards, fibre boards**
Bamboo, reeds, elephant grass	Roof, wall panels, bamboo reinforced concrete
Fibres: sisal, jute, banana	**Fibre reinforced mud blocks, mortar, roofing sheets**
Palm leaves, areca nut husk	Thatch, boards

- **Problems: moisture absorption, rot, fire, insects** – need preservative treatment, coating, cement / lime / resin binders, and **proper detailing** (raised plinth, wide eaves)
- **Fire retardant treatment of thatch:** dipping in solutions of **borax + boric acid, ammonium phosphate / sulphate, or alum**; or covering the thatch with **mud–lime slurry plaster**

8. Alternative Building Maintenance

Maintenance is the **care** that keeps a building safe and usable, avoiding expensive repair. **Low cost housing** needs **simple, cheap and community-based maintenance**, with alternative materials.

- **Preventive maintenance** is better than repair: annual inspection of roof, drains, plaster, doors; keep **roof and surface drainage** clear
- **Cracks:** repair with **cement slurry, lime-surkhi mortar, epoxy injection**, stitching with bars; **damp proofing** using **lime-based / bitumen** coats, **integral waterproofing compounds** and **brick-bat coba waterproofing** for roofs
- **Termite treatment** (borax, chemical barrier) and **wood preservatives**; **lime wash** with salt / alum in rural houses every year; **cement-free paints (lime, clay, distemper)**
- **Mud houses:** re-plaster with **mud + cow dung + lime** after every monsoon; **thatch** re-laid every 3-5 years
- **Use of community / mason training**, **Building Centres** provide material, advice
- **Durable material choice** (ferrocement, FaL-G, stone) reduces cost over life: **life cycle cost approach**; **solar passive design** reduces repair of heat cracks

9. Low Cost Infrastructure Services

A house is useful only when it has **water supply, sanitation, drainage, roads, power and waste disposal**. The cost of infrastructure is **15-30%** of the cost of housing for the poor, so low cost technology is essential here too.

Present status

- **Water:** a large share of rural households have no tap water at home; **Jal Jeevan Mission (2019)** aims at **tap water connection to every rural household**; **AMRUT** in cities; slums depend on stand posts, tankers, hand pumps
- **Sanitation: Swachh Bharat Mission (2014)** has built millions of toilets and declared India ODF (2019); but **sewerage covers only about 30-35% of urban population**; septic tanks and pit latrines are widespread; **faecal sludge management** is a new challenge

- **Solid waste and drainage:** poor collection, open drains; **Energy:** electricity (**Saubhagya**), LPG (**Ujjwala**), biomass still used by many; **roads: PMGSY** for rural roads

10. Technological Options and Low Cost Sanitation

Low cost sanitation options

Option	Features
Pour-flush (water seal) latrine with twin pit (Sulabh)	**Water seal trap** (2-3 L flush) stops smell and flies; **two leach pits**, used alternately; **compost-like** safe sludge after 2 years; **cost Rs 12,000-15,000**
Single pit latrine, VIP (ventilated improved pit)	Dry pit with vent pipe, for no-water areas
Septic tank + soak pit	For houses with a bigger water use; **tank holds sewage for 24 h**, with 2 chambers; effluent to **soak pit**; desludging every 2-3 years
Community / public toilets	In slums with **bio-digester**, shared maintenance; sanitary complex
Eco-sanitation (ecosan), composting toilet	Urine diverting, **reuse as manure**; waterless
Biogas-linked toilet (bio-digesters, DRDO bio-toilet)	Anaerobic digestion, gas for cooking, **Bio-toilet** in trains
Simplified / condominial sewerage, small bore sewer	Low cost sewer for dense area, **shallow depth, small diameter**
Decentralised wastewater treatment (DEWATS), constructed wetlands	Sustainable local treatment

Design of twin leach pits (approximate)

Pit capacity = N x R x T (N = users, R = sludge accumulation rate = 0.04 m^3/person/year (wet cleansing) or 0.03 (dry), T = pit filling time = 2 to 3 years (each pit is used alternately))

For 5 users and T = 3 yr: V = 5 x 0.04 x 3 = 0.6 m^3; adopt a circular pit of **1.0 m diameter x 1.0 m depth (0.78 m^3)**; lined with **honeycomb brickwork, cover slab, 1.0 m apart** (centre to centre, double the pit dia), at least **10 m away from a water source**.

11. Domestic Waste (Wastewater and Solid Waste)

- **Grey water** (bath, kitchen, laundry) is **60–70%** of domestic wastewater; manage by **soak pits, kitchen garden reuse, trench drains, leach pits, and root-zone treatment**
- **Drainage: open surface drains (U-shaped, lined with brick, stone, or precast cement concrete channels) with gradient 1 in 200 or steeper**, covered with slabs; **lanes with central or side drains**; **storm water drains** separate; **no stagnation** to prevent mosquito breeding
- **Solid waste: segregation at source (wet / dry), composting (pit, windrow), vermicomposting, biogas plants, recycling, community bins** and door-to-door collection (**SWM Rules 2016**); **sanitary landfill** for the residue
- **Soak pit design:** capacity for the peak flow, e.g., 135 lpcd x 5 persons x 0.3 (grey part) = about 200 L/day => pit 1 m dia x 1 m deep filled with **brick-bats / stones**, with 300 mm sand layer; **not near wells**

12. Water Supply (Low Cost Options)

Option	Details
Hand pump (India Mark II / III)	Bore well with hand pump for **250 persons per pump**, up to 50 m depth
Public stand posts	Tap, 1 tap per **100–150 persons**, in slums (40 lpcd)
Gravity flow scheme / piped scheme	From a spring or lake in hills; **cheap, no power**
Open dug well and protected well	Shallow wells with parapet, platform, cover; **chlorinated**
Rain water harvesting (RWH)	Roof catchment, **first-flush** device, filter, **storage tank or recharge pit**; **V = A x R x C** (A = roof area, R = rainfall, C = 0.8)
Household treatment	**Boiling, bleaching powder, chlorine tablets, candle (ceramic) filters, slow sand filter, solar disinfection (SODIS), bio-sand filter**
Community RO / water ATM, tankers	Urban slums

Example: rain water harvesting potential
Roof area 50 m^2, annual rainfall 700 mm, run-off coefficient 0.8. V = 50 x 0.7 x 0.8 = **28 m^3** = 28,000 litres per year (about 77 L/day)

13. Energy

- **Energy in housing:** cooking (largest in rural houses), lighting, heating / cooling, water heating, appliances; low cost solutions **cut both bills and pollution**
- **Cooking: improved smokeless chulha (cookstove)** with 25-30% efficiency vs 10% of traditional one; **biogas plant (family size 2-4 m^3 from cattle dung)**, **LPG (Ujjwala)**, **solar cookers**
- **Lighting: LED lamps (UJALA)**, **solar lanterns and home lighting systems, rooftop PV (PM Surya Ghar)**, **solar street lights**
- **Water heating: solar water heaters** (100-200 L) save electricity
- **Passive solar design / energy efficient buildings:** orientation, **cross ventilation, shading by chajjas, roof insulation (white paint, filler slab, mud phuska), thick walls and cavity / rat-trap walls, light colours, skylights**, ECBC (Energy Conservation Building Code)
- **Others: wind mill and micro hydel** in suitable areas, **community biogas, bio-mass gasifiers, solar pumps**

14. More Alternative Materials at a Glance

Material	Key facts
Stabilised soil (compressed earth) blocks – CSEB	Soil + 5–10% cement (or lime) pressed in a manual / hydraulic press (**Auram press**); size 230 x 190 x 100 mm; **strength 3–5 MPa**; no burning; **30–40% cheaper than brick**; thermal comfort
Autoclaved aerated concrete (AAC) blocks	Fly ash + lime + cement + aluminium powder, autoclaved; **density 550–650 kg/m^3**, light, insulation, fast laying with thin-joint adhesive
Cellular light weight concrete (CLC)	Foam concrete, density 400–1800 kg/m^3, panels and blocks
Micro concrete roofing (MCR) tiles	Thin cement-sand tiles 6–12 mm, **cheap roof for rural houses**, made in small units; **cost 30% less than clay tiles**
Bamboo reinforced concrete	Treated bamboo strips as steel substitute in **lintels, slabs, columns** with low loads; **bond improved with bitumen / sand coat**
Fibre cement / asbestos-free sheets, bamboo mat corrugated sheets	Light roofing
Sandwich (EPS) panels / PUF panels	Two cement skins with insulating core; quick walls
Geopolymer concrete	Fly ash + alkali activators, **no cement**, low CO_2
Plastic / recycled plastic bricks and tiles	Waste plastic bonded, low cost

- Before use, a new material must pass **tests of strength, durability, water absorption and fire**; get **BIS or BMTPC (PACS) approval**; pilot demonstration; **build skill in the community** through training at **Building Centres**

Remember for exam

* Ferrocement: thin (10–40 mm), mortar 1:2 with multiple wire mesh layers, 2–8% steel; cure 14 days
* Gypsum board: paper-faced boards on metal studs; GFRG panels for load bearing; not for wet areas
* Fly ash, slag, red mud, phospho-gypsum = industrial wastes; RHA, bagasse, straw, coir = agricultural wastes
* Bamboo, ferrocement, RCC, FRP, particle board replace timber

* Twin pit pour-flush latrine, 1 m dia pits; hand pump for 250 persons; improved chulha, biogas, solar

Important questions

1. Suggest substitutes for scarce building materials such as cement, steel and timber.
2. What is ferrocement? Explain its materials, construction and applications.
3. Write a note on gypsum boards and their use in low cost housing.
4. Explain the alternatives to timber in doors, windows and roofs.
5. Discuss the use of industrial wastes and agricultural wastes as building materials.
6. Explain alternative building maintenance for low cost houses.
7. Describe the present status and technological options of low cost infrastructure services.
8. Explain low cost sanitation options. Design a twin pit latrine.
9. Discuss low cost water supply and energy options for housing.

Cost Effective Housing Techniques

Unit IV: Rural Housing

Syllabus

Rural housing: introduction - traditional practice of rural housing - mud housing technology - mud roofs - characteristics of mud - fire treatment for thatch roof - soil stabilization - rural housing programme.

1. Introduction

Nearly **two-thirds of Indians live in about 6.5 lakh villages**. A rural house is **home, workplace (farm, craft, animals), store and social space** together. Rural housing is built by the family itself, with **local materials** (mud, thatch, stone, bamboo, timber) and **traditional skills**. The main need is to **improve** these houses - for durability, safety and hygiene - **without losing the cost advantage**.

Features of rural housing

- **Self-help construction** with village masons, using **local materials**, little cash
- **Climate responsive** design: thick mud walls and thatch keep the house cool in summer and warm in winter
- House combined with **cattle shed, grain storage, courtyard (angan), work area**
- Built **incrementally** (room by room as money allows)
- **Problems:** poor durability, **damage from rain, flood, fire, wind and earthquake**, termite attack, smoke from cooking, no toilets, low ventilation and light, overcrowding

Aspect	Traditional rural house	Needed improvement
Walls	Mud, wattle-and-daub, stone, bamboo	Plinth protection, stabilised mud, bands
Roof	Thatch, mud terrace, country tiles	Fire-proof and leak-proof, bamboo / MCR tiles, RCC / precast
Floor	Mud / cow dung	Cement / brick floor with damp proof
Cooking	Chulha in room, smoke	Smokeless chulha with chimney, LPG, kitchen vent
Sanitation	Open defecation	Twin-pit latrine, bath area
Water	Well, pond	Hand pump / tap, RWH

2. Traditional Practice of Rural Housing

Every region has developed a house type suited to its **climate, materials and way of life**.

Region	Traditional house
Rajasthan, Gujarat (arid)	**Mud / adobe walls, flat mud roofs (terrace), circular "Bhunga" huts** of Kutch (cyclone and earthquake resistant), thick walls, small openings, lime wash
Kashmir (cold, seismic)	**Dhajji-dewari** (timber frame with brick or stone infill), sloping roofs for snow; **taq** system (timber-laced masonry)
Himalayan states (Uttarakhand, HP)	**Stone masonry with timber bands, slate roofs, Kath-kuni** (timber-stone cribbing)
North East (Assam, Manipur)	**Ikra (bamboo reed plastered with mud) walls, Assam-type houses, stilt houses (chang ghar)** against floods and animals, bamboo and thatch roofs
Kerala, coastal belts	Laterite stone and timber, **steep tiled / thatch roofs** (to drain heavy rain), wide verandas
Indo-gangetic plains (UP, Bihar, Bengal)	**Mud / mud-brick walls, thatch or country-tile (khaprail) roofs**, bamboo; plinth raised against floods; Bengal **"Bangla" (curved) roofs**
Deccan plateau and South	Stone / mud walls, **Madras terrace and tile roofs**, central courtyards
Tribal areas	Round huts of bamboo, mud, grass; shifting dwellings

- **Lessons:** use of **locally available materials**, **steep roofs in heavy rain**, **stilts in flood**, **thick walls in hot-dry**, **light and flexible structures (bamboo, timber) in earthquake**, **low height** and **compact plan** for strong winds, **courtyard** for light and ventilation

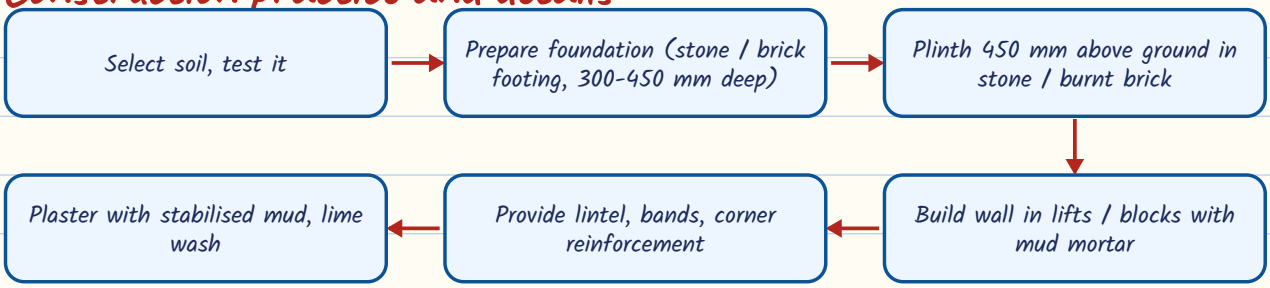
3. Mud Housing Technology

Earth (mud) is the oldest and the cheapest building material, available at site for free. Earthen buildings are **cool in summer, warm in winter**, need no fuel to make, are fully **recyclable**, and have **very low embodied energy**. Weakness: low strength and **erosion by water**.

Methods of mud construction

Method	Description
Cob (monolithic mud walls)	Wet soil mixed with **straw / husk**, built up in **layers (lifts) of 300–600 mm height**, allowed to dry (a day) before the next; wall thickness **300–600 mm**, trimmed after drying
Adobe (sun-dried mud blocks)	Soil + straw moulded in wooden moulds (e.g. **300 x 150 x 100 mm**), **sun dried for 14 days**, laid in **mud mortar**
Rammed earth (pise)	Moist soil (**about 10% water**) **compacted in layers of 100–150 mm** in wooden formwork; strong (**1.5–3 MPa**) and durable; thickness 300–450 mm
Wattle and daub (ikra, bahareque)	Bamboo / wood frame, woven mat (**wattle**), plastered with **mud and cow dung (daub)**; **light and earthquake resistant**
Compressed stabilised earth blocks (CSEB)	Soil + cement / lime pressed in a machine; stronger and uniform
Mud-brick (kachcha brick)	Moulded and sun dried, like adobe

Construction practice and details

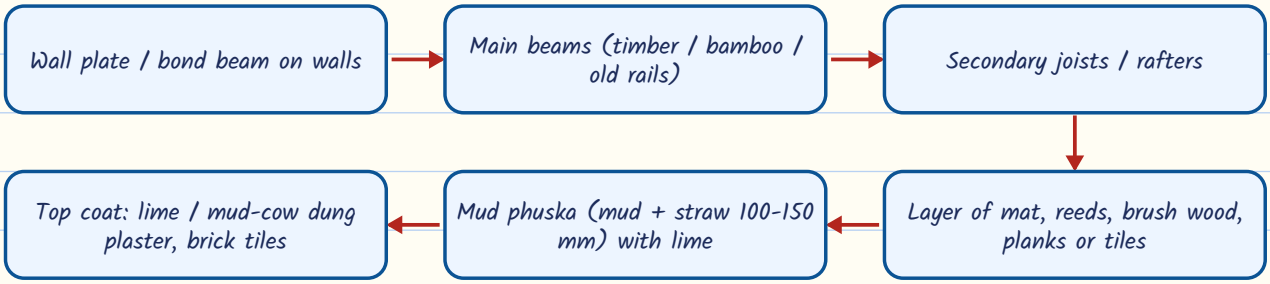


- **Foundation and plinth: stone or burnt brick footing** (not mud) at least **300–450 mm** deep and a **plinth of 450 mm** with a **damp-proof layer** (stone slab / plastic sheet) so that capillary water does not damage the mud wall
- **Wall thickness: not less than 300 mm** (cob 450–600 mm); **height max 8 times thickness**, i.e., not more than **2.7–3 m (single storey)**; **length of wall between cross walls max 6 m**; **openings** small, **not more than 1/3** of wall length, **at least 1 m from corners**
- **Lintels** of wood, bamboo or **RCC**, **bands** of timber / bamboo / RCC at **lintel and eave level** (earthquake safety); **corner reinforcement** (bamboo / steel / geogrid)
- **Protection of walls: wide roof overhang (min 750 mm)**, **plaster** (mud + straw + cow dung, **lime-cement**), **bitumen or lime wash** (coatings), **stone aprons**, **gutters and drains** away from walls, **no leaking taps and pipes**

- **Disadvantages:** erodes in rain and floods, **low strength**, termites, cracks, **needs yearly maintenance**; poor image

4. Mud Roofs

Mud roofs **(flat or low pitched terraces)** are common in **dry and semi-arid areas (Rajasthan, Gujarat, MP, Deccan)**, where rainfall is low.



- **Mud phuska roof:** soil mixed with **chopped straw / husk**, laid in 100-150 mm and well rammed, with **slope 1 in 30 to 1 in 20**; finished with **mud plaster and a layer of lime or bitumen**; **repaired every year before rains**; **weight 250-300 kg/m^2** (heavy, so **bad for earthquakes**)
- **Improved mud roof: waterproofing** layer **(polythene sheet / bitumen coated felt / lime-brick-bat layer)**, **good drainage with spouts**, **parapet 150 mm**, **stabilised mud**, **roof bands**; **thermal comfort is excellent**, because mud has a high heat capacity and low conductivity
- **Other roofs: thatch (see below)**, **country tiles (khaprail) on bamboo / timber rafters**, **Mangalore tiles**, **MCR tiles**, **GI / AC / bamboo mat sheets**, **Madras terrace**, **brick-on-edge arches**, **precast plank-and-joist**, **filler slab**

Design points for pitched roofs in rural areas

* **Slope 30 - 45 degrees** for tiles, **45 - 60 degrees** for thatch (rapid water shed)
* **Overhang (eaves) 600-900 mm**, rafters tied; **hip roofs** better in cyclone areas than gable roofs
* Light roof (tiles / thatch / sheets) for **earthquake zones**; **tie roof to wall** firmly **(anchor bolts, wire ties)**

5. Characteristics of Mud (Soil as a Building Material)

Soil is a mixture of **gravel (above 2 mm)**, **sand (0.06–2 mm)**, **silt (0.002–0.06 mm)** **and clay (below 0.002 mm)**. Clay is the **binder**, sand/gravel the **filler / skeleton**.

Property	Desirable value / meaning
Composition for mud walls	**Clay 15 – 25%, silt 10 – 30%, sand 50 – 70%, no organic matter**
Compressive strength	**0.5 – 3 MPa** (dry), depends on clay, compaction and moisture; **wet strength much lower**
Plasticity	**Liquid limit 25 – 50%, plasticity index 7 – 18** - too much clay cracks, too little crumbles
Shrinkage	**Clay shrinks 5–10% on drying: cracks**; reduce by adding **sand / fibre**
Density and thermal property	**1600 – 2000 kg/m^3**; thermal conductivity **0.5 – 0.9 W/mK** (good insulation), high heat capacity (**thermal lag**)
Water absorption and erosion	High: **must be protected**; swelling in clay
Fire and sound	Fireproof; good sound insulation

Field tests of soil

Test	Procedure	Meaning
Ribbon (roll) test	Roll a moist ball into a thread of 3 mm dia; 150–250 mm length without breaking	Good clay content
Dry strength / cigar test	Make a ball of 20 mm, dry; try to crush	Resists = good binder
Jar (sedimentation) test	Shake soil with water in a transparent jar; read layers after 1 and 24 h	Percentage of sand, silt, clay
Shrinkage box test	Fill a **60 x 60 x 600 mm** box with moist soil, dry, measure shrinkage	Cracks and length decrease should be below 15–20 mm
Drop (ball) test	Drop a 40 mm ball from 1.5 m	Breaks into few pieces = OK
Smell and visual test	Musty smell shows organic matter, which is bad	Reject soil with humus

6. Soil Stabilization

Soil stabilisation improves the **strength, water resistance (durability)** and reduces **shrinkage and swelling** of the soil by mechanical, chemical or physical means.

Stabiliser	Dosage	Best for / effect
Compaction (mechanical)	Rammed layers, compaction **at optimum moisture**	Cheap, increases density and strength
Cement	**5 – 10%** by weight (**sandy soils, PI below 15**)	Hydration gives bonding, high strength (3-5 MPa), reduces swelling; cure for 7-28 days
Lime	**6 – 12%** (**clayey soils, PI above 15**)	Reacts with clay (pozzolanic): **strength, less plasticity**; slow
Bitumen / cut-back / emulsion	**4 – 6%** (sandy, low clay)	**Water repellent** (waterproofing); no strength gain
Fly ash, rice husk ash, slag (with lime)	10-30%	Pozzolanic, cheap
Fibres: straw, jute, coir, sisal, bamboo chips	**0.5 – 2% by weight**	**Reduce cracking, improve tensile strength**
Cow dung, molasses, gum, tannin	Plaster and wash coats	Traditional, water resistant
Gypsum, calcium chloride, enzymes, polymers, resins	Small %	Special / modern chemical stabilisers

Rules of thumb

* **Sandy soils**: use cement or bitumen; **clayey soils**: use lime (or lime + cement), fibres
* **Cement-soil blocks**: soil (sieved) + 6-8% cement, water to optimum moisture content, **press, cure for 14 days under shade** (keep damp)
* **Stabilised soil must be tested**: wet compressive strength not less than **1.5-2 MPa**; water absorption max 15-20%

Example: mix for 1000 stabilised blocks

Block size 230 x 190 x 100 mm = 0.00437 m^3; compacted density 1900 kg/m^3 => 8.3 kg per block => 8300 kg soil mix per 1000 blocks.

With 7% cement: cement = 0.07/1.07 x 8300 = **543 kg (11 bags)** and dry soil = 7757 kg

Cost advantage = about 30% lower than burnt bricks (no fuel).

7. Fire Treatment for Thatch Roof

Thatch (paddy straw, wheat straw, grass, reed, palm leaves, coconut leaves) is cheap, light and an excellent **insulator**, but is very **flammable** and is attacked by rain, insects and fungus; **life of 5-10 years**. **Fire** is the biggest danger in villages (cooking fires, sparks, short circuits).

Fire retardant treatments

Treatment	Method
Mud-lime or mud-cement slurry coating	Spray / apply a plaster of **mud + cow dung + lime / cement** on the thatch surface (5-10 mm)
Chemical solutions (retardants)	**Borax + boric acid** (ratio 1:1), **ammonium sulphate / phosphate, alum, sodium silicate, zinc chloride**: a solution of about **10-15%** is sprayed or the thatch is **dipped** and dried
Lime wash / Cement-lime paint	Over the top
Thatch over fire-proof underlayer	Lay **bamboo mat plastered with mud**, **or a tarred / GI sheet / bricks** beneath
Use of fire barrier	Keep thatch house **at least 5-10 m apart** and **chulha away** from thatch, **chimney with spark arrestor**, **separate kitchen** with **fire resistant roof** (tiles, sheets)

- **Other treatments: neem oil or copper sulphate for fungus and insects**; **bitumen emulsion for waterproofing**; **steep slope of 45-60 degrees**, thickness **150-250 mm**, **tied** to bamboo battens by coir / wire; **replace damaged thatch** before monsoon
- **Fire safety practices: water pots / pits in the village**, community firefighting awareness, **electrical wiring with proper insulation**, **smokeless chulha with chimney**, **avoid open flame lamps near thatch**

8. Rural Housing Programmes

Programme	Details
Indira Awaas Yojana (IAY, 1985-2016)	Grants for **BPL families (SC/ST, freed bonded labour priority)**, 40% quota to SC/ST; started as part of RLEGP / JRY
Pradhan Mantri Awaas Yojana - Gramin (PMAY-G, 2016)	**Housing for All in rural areas**; target **2.95 crore houses**; beneficiaries from **SECC 2011** data (kutcha and houseless); **assistance Rs 1.20 lakh (plains) and Rs 1.30 lakh (hills, NE, difficult areas)**; **minimum house size 25 sq.m**, with a **cooking space**; **toilet** via **SBM-G (Rs 12,000)** and **90-95 days MGNREGA wages**; **loan up to Rs 70,000** optional; **payment through DBT**; **Geo-tagging (AwaasApp)**, **Rural Mason Training**
Other state schemes	Samagra Awaas, Mukhyamantri Gramin Awas, Biju Pucca Ghar, Indiramma houses (AP)
Rural Housing Fund (NHB), HUDCO, NABARD	Credit for rural housing
Rural Building Centres, Nirmithi Kendras	Technology and training
Convergence	Jal Jeevan Mission, Ujjwala, Saubhagya, **MGNREGS**, PMGSY

- **Objectives:** remove **kutcha houses**, ensure **safe, durable, hygienic** houses with **basic services**; **empower the women (houses in the name of women / jointly)**; **local typology designs** (regionally suitable) given to beneficiaries; **disaster resistant features** are encouraged
- **Implementation: gram sabha** selects beneficiaries, funds in **instalments (e.g., 25%, 60%, 15%)** linked to **plinth, lintel, completion** stages, with **photo geo-tagging**; **technical support from master masons, rural mason training (RMT)** - community and **NGOs**
- **Issues:** delays, low unit assistance versus the cost of material, **rising cost of materials**, lack of technical guidance, poor quality, **land titles for landless**, **non-engineered building**, difficult hilly regions; **solutions: cost-effective technology (rat-trap, stabilised blocks, filler slab, MCR tiles), housing finance, training, monitoring, convergence**

Typical low cost rural house design (about 25-30 sq.m)

* Plan: **one room (3 x 3 m) + kitchen + verandah + toilet**, extendable by adding rooms **(core house concept)**

* Foundation: **stone or brick masonry in 1:8 mortar**; **plinth 450 mm**; walls: **stabilised mud blocks or fly ash bricks in 1:6 mortar (rat-trap bond), 230 mm**
* Roof: **MCR tiles / GI sheets on bamboo or steel trusses, or filler slab / precast plank-joist**, with **0.75 m overhang**
* **Smokeless chulha**, **twin-pit latrine**, rain water harvesting, **cost about Rs 1.2 – 2.5 lakh** (varies)

Design considerations for rural houses

- **Climate-responsive** orientation (long axis east-west so that the long walls face north and south; small openings on the west), **cross-ventilation**, **shading**, **thermal comfort**
- **Disaster-resistance** (flood, cyclone, earthquake, fire) as per **IS 4326, IS 13827 (earthen buildings in seismic zones), IS 13828 (low-strength masonry), IS 15883 (guidelines for construction)**
- **Social needs: cattle shed separated**, **grain store**, **courtyard**, **women's working space**, **toilet attached**

9. Bamboo in Rural Housing

- Bamboo is abundant in **NE India, Kerala, Bengal, Odisha**; grows to maturity in **3–5 years**; **tensile strength about 100–300 MPa**; light and flexible, so **good for earthquakes**
- **Uses: posts, rafters, trusses, wall frames (wattle), mats (bamboo mat board, ikra), bamboo reinforced concrete, scaffolding, floor, doors / windows**
- **Treatment is a must: borax-boric acid (1:1.5), 5% solution by the "Boucherie / soak and diffusion" method for 7–10 days**, or **smoking**, or **chemical-pressure treatment**; protect from **moisture** with a raised footing on a stone or concrete base (never bury bamboo in the ground)
- **Joints** are the weak point: use **bolts, lashing with GI wire / coir rope, steel brackets, and mortar filling inside bamboo node**
- **Bamboo mat corrugated sheets (BMCS)** are a cheap roofing; **Bamboo mat board** in partitions

10. Flood, Cyclone and Fire Safe Features in Rural Houses

Hazard	Features
Flood	**Plinth above high flood level (HFL)**, stilt / platform house, **stabilised mud or burnt brick plinth**, water-resistant plaster, **raised platform (community)**, **no mud walls below flood level**
Cyclone / strong wind	**Hip roof, slope 22–30 degrees**, low height (single storey), **anchor roof to wall by ties and bolts**, **small overhang**, strong foundation, **no heavy tiles unattached**, shelters
Fire	**Fire-retarded thatch, separate kitchen, brick / stone chimney, spacing 5–10 m**
Earthquake	**Light roof, symmetrical plan, bands, corner reinforcement**, no heavy mud roofs
Termite and rain	**Termite barrier (soil treatment), wide eaves, drainage around the house, lime wash**

11. Annual Maintenance Calendar of a Mud House

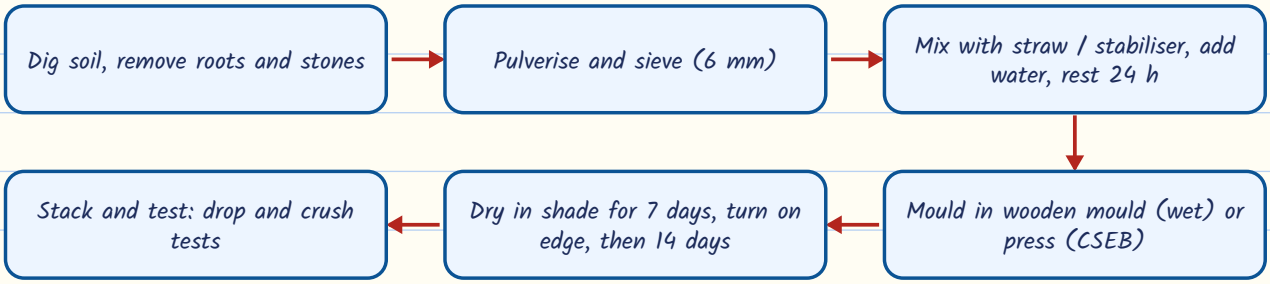
Time	Work
Before monsoon (April–May)	Replaster walls with mud-cow dung (or stabilised mud); repair cracks; repair roof (phuska / thatch / tiles); clear drains and gutters
During monsoon	Inspect leaks, keep surroundings dry, protect plinth, remove vegetation near walls
After monsoon (Oct–Nov)	**Lime wash** with alum / salt; fill cracks with mud slurry; termite check
Every 3–5 years	Re-lay thatch; replace rotten timber / bamboo; strengthen foundation aprons
Every 10 years	Replace damaged mud blocks, repair the plinth, re-do major finishing

12. Cost Estimate of a Core Rural House (Illustrative)

Item	Approximate share of cost
Foundation and plinth	15%
Walls (stabilised mud / fly ash bricks)	30%
Roof (MCR tiles / sheets / precast)	25%
Doors, windows, flooring	12%
Plaster, painting, finishing	8%
Toilet, water, electricity	10%

Total for a 25 sq.m house is about **Rs 1.5 – 2.5 lakh** (as per local rates); **PMAY-G assistance (Rs 1.2 / 1.3 lakh)** + **MGNREGS wages** + beneficiary contribution (own labour) fill the cost gap. Use of **local labour and materials** cuts it by 20-30%.

13. Production of Adobe and Stabilised Blocks (Step by Step)



- **Quality checks:** no cracks, sharp edges, **dry compressive strength min 2.5-3.5 MPa**, wet strength min 1.5 MPa after 24 h soaking, and **erosion test (spray of water 1 h under 0.5 m head)** with depth of pitting less than 15 mm
- **Mud mortar:** same soil as block, with **more sand and fibre**, thin joints (10-12 mm), **fill all joints**; **cement-mud mortar 1:12** for improved strength
- **Quantity:** a 3 m x 3 m room with 300 mm walls (2.7 m high) needs about **1100-1300 adobe blocks** (300 x 150 x 100 mm)

14. Rural Building Centres and Technology Transfer

- **Rural Building Centres (RBCs)** and **Nirmithi Kendras** (Kerala), **HUDCO building centres**, **KVKs**, NGOs like **Auroville Earth Institute, Hunnarshala (Gujarat), Laurie Baker Centre (COSTFORD)** train masons, show **demonstration houses**, and **produce stabilised blocks, tiles, door and window frames**
- **Rural Mason Training (RMT)** under PMAY-G certifies masons for **disaster-resistant and cost-effective construction**
- **Role of institutions: CBRI Roorkee** (mud and low cost technologies), **SERC Chennai**, **BMTPC** (guidelines and Vulnerability Atlas), **NIRD & PR Hyderabad** (training for rural development), **IITs and NITs**
- **Participatory approach:** involve **women, SHGs and gram panchayat**; **owner-driven reconstruction** (post-disaster Gujarat, Maharashtra) gave good results

Remember for exam

* Mud house types: cob, adobe, rammed earth, wattle-and-daub; wall thickness not less than 300 mm; plinth 450 mm
* Good mud soil: sand 50-70%, silt 10-30%, clay 15-25%; tests: ribbon, jar, shrinkage box, drop test
* Stabilisers: cement 5-10% for sandy soil, lime 6-12% for clayey soil, bitumen 4-6% for waterproofing, fibres 0.5-2%
* Thatch fire retardants: borax-boric acid, ammonium sulphate / phosphate, alum, mud-lime coating
* PMAY-G: 2.95 crore houses, Rs 1.2 lakh (plains) / 1.3 lakh (hills), 25 sq.m with cooking space, SECC data, geo-tagging

Important questions

1. Describe the traditional practices of rural housing in different regions of India.
2. Explain the mud housing technology: cob, adobe, rammed earth and wattle and daub.
3. Discuss the construction of mud roofs and their maintenance.
4. Explain the characteristics of mud. How is soil tested for use in mud construction?
5. What is soil stabilisation? Explain the methods with the stabilisers and their dosage.
6. Explain the fire treatment for thatch roofs.
7. Describe the rural housing programmes of the government (IAY / PMAY-G).

8. Design a low cost rural house with suitable materials and technology.

Cost Effective Housing Techniques

Unit V: Housing in Disaster Prone Areas

Syllabus

Housing in disaster prone areas: introduction - earthquake - damages to houses - traditional / disaster prone areas - types of damages and failures of non-engineered buildings - repair and restoration of earthquake damaged non-engineered buildings - recommendations for future constructions - requirements of structural safety of thin precast roofing units against earthquake forces - status of R&D; in earthquake strengthening measures - floods, cyclone, future safety.

1. Introduction

India is one of the **most disaster-prone countries**. About **59% of the land is earthquake prone, 12% flood prone (40 million hectares), 8% cyclone prone** (7500 km coastline) and 68% of the area is drought prone. Disasters kill thousands, destroy lakhs of houses and push the poor back into poverty. **Houses (especially non-engineered) are the main source of loss of life**, so **safe housing** is the best disaster protection: "**Earthquakes do not kill people, buildings do.**"

Hazard	Housing impact
Earthquake	Collapse of walls and roofs, killing people inside
Flood	Submergence, erosion of foundations, collapse of mud walls
Cyclone / storm surge	Roof blown off, wall collapse, flooding of coastal houses
Landslide	Burial of houses in hills
Fire, drought, tsunami (2004)	Loss of houses, services, livelihoods

- **Disaster management cycle:** prevention - mitigation - preparedness - response - relief - **recovery (reconstruction)** - development; **Disaster Management Act 2005**, **NDMA** (National Disaster Management Authority) and **State SDMAs**; "**Build Back Better**" is the approach for reconstruction
- **Vulnerability Atlas of India (BMTPC, 1997, 2006, 2019)** shows **hazard-wise vulnerability of housing for each district** (wind, flood, seismic, landslide) and the

number of houses at risk

2. Earthquake: Basic Concepts

An **earthquake** is sudden shaking of the ground caused by the release of **strain energy** stored in rocks (along **faults**, plate boundaries). The Indian plate moves north and collides with the Eurasian plate: it causes the **Himalayan** earthquakes.

Term	Meaning
Focus (hypocentre)	Point under the ground where rupture starts
Epicentre	Point on the surface directly above the focus
Magnitude	Energy released (**Richter scale**, log scale; each unit = 32 times energy)
Intensity	Effect on structures and people at a place (**MSK / Modified Mercalli, 12 grades**); decreases with distance
Seismic waves	**P waves (compression, fast), S waves (shear, damaging), surface waves (Love, Rayleigh)**
Seismograph / accelerograph	Records ground motion; **PGA (peak ground acceleration)**
Liquefaction	Loose saturated sand loses strength: buildings tilt and sink
Tsunami	Sea waves due to sea-bed earthquake

Seismic zones of India (IS 1893 (Part 1): 2016)

Zone	Zone factor Z	Severity	Areas (examples)
V	0.36	Very severe	**Kashmir, NE India, Rann of Kutch, Andaman**
IV	0.24	Severe	**Delhi, Bihar, Uttarakhand, Sikkim, parts of Gujarat and Maharashtra coast**
III	0.16	Moderate	**Mumbai, Chennai, Kerala, parts of WB, Odisha, Rajasthan**
II	0.10	Low	**Most of the peninsular India: Hyderabad, Bengaluru (in II)**

- Zone VI was removed from the 2002 and later revisions; **zone map has only 4 zones (II to V)**
- **Notable earthquakes: Latur (Killari) 1993 (M 6.2), Uttarkashi 1991, Chamoli 1999, Bhuj 2001 (M 7.7, 13,800 deaths), Kashmir 2005 (M 7.6), Sikkim 2011, Nepal 2015** - most damage to **non-engineered** masonry and mud houses

Seismic design force (IS 1893)

Design horizontal seismic coefficient: $A_h = (Z/2) \times (I/R) \times (S_a/g)$

Design base shear: $V_B = A_h \times W$ (W = seismic weight of the building)

Z = zone factor, I = importance factor (1.0 / 1.5), R = response reduction factor (1.5 for unreinforced masonry, 3-5 for RC frames), S_a/g = spectral acceleration

Example

A one-storey masonry house, W = 600 kN, Zone IV (Z = 0.24), I = 1, R = 1.5, S_a/g = 2.5.

$A_h = (0.24/2) \times (1/1.5) \times 2.5 = 0.20$; $V_B = 0.20 \times 600 =$ **120 kN** (20% of the weight acts horizontally!) - hence **light roofs** are important.

3. Damages to Houses in Earthquakes

Earthquake force is an **inertia force = mass x acceleration**: heavier building, higher force. Walls and roofs shaken **in-plane and out-of-plane**. **Non-engineered buildings** (built without engineer, by owner or village mason with local materials: **mud, adobe, stone, brick in mud, low-strength masonry**) form **majority of Indian housing** and cause the maximum deaths.

Disaster-prone areas (regional vulnerability)

Region / Hazard	Typical vulnerable houses
Himalayan belt, NE India (seismic IV-V)	**Stone and mud masonry, brick in mud, heavy slate / mud roofs**
Gujarat, Kutch	Stone and brick houses, **heavy RCC / stone slab roofs**, bhungas (good)
Indo-Gangetic plain (Bihar, UP) - seismic and flood	**Mud / adobe, kachcha houses**
East and West coasts - cyclone, storm surge	Kachcha, thatch, weak masonry
Brahmaputra and Ganga basins - floods	Mud and bamboo houses
Central India (Latur) - moderate seismic but very vulnerable houses	**Heavy stone masonry with mud mortar and heavy mud roofs**

4. Types of Damages and Failures of Non-Engineered Buildings

Failure type	Description and cause
Diagonal (X) shear cracks in walls	Due to **in-plane shear**; around openings; weak mortar, long walls
Separation of walls at corners and T-junctions	Walls vibrate in different directions; no **corner stitching / bands**; vertical crack at corners
Out-of-plane collapse of walls (overturning)	Walls too tall, thin, slender, unsupported by cross walls or roof; **mud mortar**
Gable wall failure	Triangular gable not tied to roof
Collapse of roof / floor	**Heavy roofs (mud phuska, stone slab) fall; no connection of roof to walls; insufficient bearing**
Delamination of walls (wythes separation)	Two-leaf stone walls, no through stones
Damage around openings	Large openings near corners, **no lintel band**, door-window cracks
Parapet, chimney, balcony, water tank, cornice falls	Cantilevers and projections
Foundation failure, settlement, liquefaction	Poor soil, no plinth band; tilted houses (Bhuj, Kutch)
Pounding between buildings	Adjacent buildings of different heights without gap
Soft storey, short column, torsion (RC buildings)	Open ground floor for parking, irregular plan
Shattering of precast roofing / slippage of units	Inadequate bearing, no connection

Causes of poor performance

* **Heavy roof** and walls (large mass), **weak materials** (mud, lime mortar, random rubble, poor bricks), **poor workmanship**
* **No horizontal bands, no vertical bars at corners, long slender walls**, unsymmetrical plan, **big openings**
* **Lack of enforcement of codes**, no engineers, **ignorance**, and **weak connection of parts** (wall-wall, wall-roof)

5. Repair and Restoration of Earthquake-Damaged Non-Engineered Buildings

After the earthquake, a **damage assessment** by trained teams classifies the buildings (e.g., **Green – safe, Yellow – restricted use, Red – unsafe**) before deciding on **demolition, repair or strengthening**.

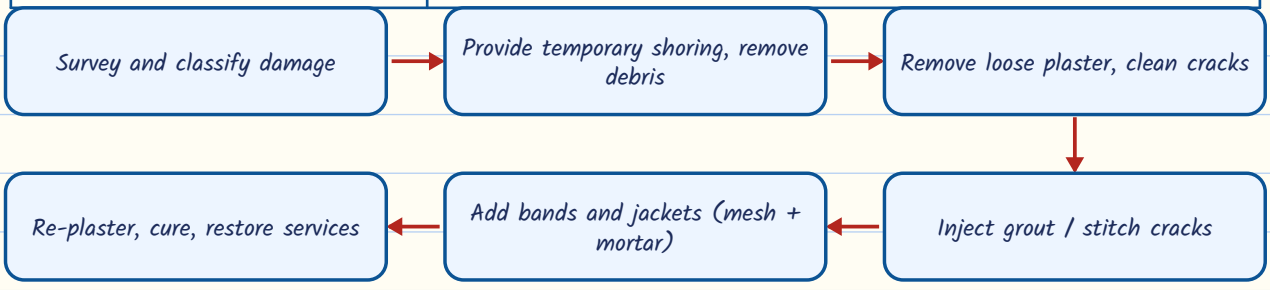
Damage grade (EMS)	Condition	Action
G1 Negligible / slight	Hairline cracks, small plaster falls	Cosmetic repair
G2 Moderate	Cracks in many walls, some plaster fall	**Repair + strengthening (non-structural / structural)**
G3 Substantial / heavy	Large wide cracks, wall failure, tilting	**Major structural repair and retrofit**
G4 Very heavy	Serious wall failure, partial roof collapse	Demolish or major reconstruction
G5 Destruction	Total or near collapse	Demolish and **rebuild** (with seismic design)

Terms

- **Restoration (repair):** bring the damaged building back to the **original strength**
- **Seismic strengthening / retrofitting:** upgrade strength **beyond the original** by adding new elements to resist future earthquakes
- **Rehabilitation:** repair + strengthening + services restored; **reconstruction:** new construction after demolition

Repair techniques (IS 13935: Repair and seismic strengthening of buildings)

Technique	Description
Crack filling / grouting	Clean cracks; **cement-sand grout / lime grout / epoxy injection**; **cement slurry for cracks 3-5 mm**, epoxy resin for cracks less than 3 mm; hairline cracks by **cement paste**
Stitching of wide cracks	**U-shaped steel dowels / stitching bars (6-8 mm) in grooves** across cracks, grouted with cement mortar
Re-building of badly damaged wall portions	Dismantle and re-build with **cement mortar**, **toothed joints**
Splints and bandages	**Reinforced concrete (or ferrocement) belts / bands** around the house at plinth, sill, lintel and eaves levels, with **vertical splints** at corners and junctions (bars in grooves, then plaster) **(2 or 3 per wall)**
Jacketing (ferrocement / mesh with mortar)	**Welded wire mesh (50 x 50 mm, 3-4 mm wire) fixed on both faces of wall by cross-ties through the wall, then plastered with 1:3 cement mortar (30-40 mm)**; **very effective and cheap**
Shotcrete / gunite overlay	Sprayed concrete over mesh
Grouting of walls and replacement of mortar	**Injection of cement-sand or lime grout** to strengthen weak rubble walls; **re-pointing**
Strengthening of foundation, roof connections	**Anchoring the roof** to walls through **bolts, wire ties; replace the heavy roof by a light roof**
Strengthening of openings	**Frames of steel angles** around big openings; fill unneeded openings
Steel belts / tie rods / buttresses	External supports, **corner buttresses**
Fibre reinforced polymers (FRP) wrapping	Carbon / glass fibre strips; higher cost



- **Precautions: materials (cement mortar not mud), hold-fasts / cross ties spaced 500-600 mm**, curing 7-14 days, **work by trained masons**, use of **local materials**, **owner participation**; **do not re-use the broken heavy roof**

- **Cost:** repair and strengthening usually **30–50% (or 10–30%)** of reconstruction cost; saves resources and time

6. Recommendations for Future Constructions

IS 4326:2013 (Earthquake resistant design and construction of buildings: Code of practice), **IS 13827** (Earthen buildings), **IS 13828** (Low strength masonry buildings), **IS 13920** (Ductile detailing of RC), **IS 1893** (design criteria), **IS 15988** (assessment and retrofitting of RC buildings) and **NDMA guidelines**.

General principles

- **Simple, symmetrical, regular plan** (square or rectangular, **L / B not more than 3**); avoid **L, T, U shapes** - separate them by **seismic joints (gap 25–40 mm)**
- **Light roof and light walls**; **low height** (1–2 storeys for masonry; **storey height 2.7–3.2 m**; **mud buildings single storey** only)
- **Box action:** all walls and roof **tied together** by **horizontal bands and vertical bars**
- **Good foundation** on uniform firm soil; **no foundation on liquefiable / filled soil**; plinth band
- **Quality of material and workmanship**, **cement or cement–lime mortar (1:6)** in zones III–V; **proper bonding of masonry (header and stretcher)**, **through stones** in stone masonry (every 1.2 m), **toothed joints**

Provision (IS 4326 / 13828 / 13827)	Requirement
Wall thickness	Brick: min **230 mm**; stone: **350–450 mm**; **earthen walls: 300 mm minimum**
Mortar	Cement-sand **1:6** (zones III–IV), **1:4** (zone V) for brick; lime-cement mud only for low-rise **earthen buildings with bands**
Openings	**Total width not more than 50% of wall length (single storey), 42% (two storey)**; **min 600 mm from corner**, pier width not less than **340 mm**, **door–window up to 1/3 length in low strength masonry**
Horizontal bands	**Plinth, lintel, sill (optional), roof / eaves (gable) level**; RCC band **75–150 mm thick, 2 bars of 8–12 mm**, stirrups 6 mm @ 150 mm; **timber or bamboo bands in low cost houses**
Vertical bars	**At corners and T junctions**, jambs of doors: **8–12 mm bars (or bamboo) from foundation to roof**, in zone IV–V
Wall height / length	Wall height **max 8 x thickness (mud: 6–8 x)**; **length between cross walls max 6 m (4 m for mud)**

Provision (IS 4326 / 13828 / 13827)	Requirement
Roof	**Light** (GI sheets, tiles, thatch on timber / bamboo trusses), **anchored with ties**; avoid heavy mud / stone slabs
Plinth and foundation	Depth 450–750 mm, width 600–900 mm in stone / brick; **continuous**

Seismic features of a low-rise masonry house

* **Roof band + lintel band + plinth band** make the house a rigid box
* **Corner reinforcement (steel / bamboo)** and **jamb bars, gable band**
* **Light roof tied** with bolts; **small openings**; **symmetry**; **mortar strong**

7. Safety of Thin Precast Roofing Units against Earthquake Forces

Thin **precast roofing / flooring units** (planks, joists, channels, cored units, ferrocement elements) are popular for low cost housing (Unit II), but they **can fall off** the walls in earthquake if not **connected** properly. The roof must work as a **rigid horizontal diaphragm**, transferring the **inertia forces** to the walls.

Requirement	Details
Adequate bearing	**At least 75 – 100 mm** on the supporting wall / bond beam (units 3 m or less span); prevents **slipping and falling**
Connection of units to supports	**Dowels, anchor bars, cast-in-situ RCC bond beam on the walls**; **hooks and bolts**; mortar bed 12 mm
Integral action of units	**Shear keys between adjacent units**, filling the joints with **1:3 mortar / micro-concrete, cast-in-situ screed (40–50 mm) with welded mesh or 6 mm bars @ 150–200 mm** acts as a **diaphragm**
Ring (bond) beam at roof level	**RCC / reinforced brick band all round**, connected to the wall steel; ties for all units
Light weight	Units, screed and finishes should be light (reduces the seismic mass and lateral force)
In-plane shear capacity	Joints and screed must carry the horizontal force $H = A_h \times W_{roof}$ through the **shear friction** to the wall bands
Vertical acceleration check	Units must resist **uplift (2/3 of horizontal)**, secure with ties or dowels
Stability of individual units	Ribs / joists should not tilt: **strutting and cross ties** before screed
Supports, openings	No large cut-outs without trimmer beams; **edges stiffened**

Requirement	Details
Quality control	Concrete cube strength **M20 or higher**, proof test of units, **IS 13990 (Precast RC planks and joists for roofing and flooring)**, **IS 15916 (precast buildings)**

Check for the roof connection

Horizontal seismic force on roof = A_h W. For roof weight 150 kN, A_h = 0.15 => H = 22.5 kN to be carried through the perimeter band to the walls. With roof perimeter = 30 m, shear per metre = 0.75 kN/m: dowels and screed should be designed for this shear (easily taken by a **mesh reinforced screed** and the **bond beam**).

8. Status of R&D; in Earthquake Strengthening Measures

- **Shake table tests** of **full-scale masonry houses** (CBRI Roorkee, **IIT Roorkee, IIT Kanpur, IIT Madras, SERC Ghaziabad**) show the effect of **bands, vertical bars, bamboo / timber reinforcement and mesh jackets**
- **Retrofitting techniques** developed: **ferrocement / wire mesh jackets, shotcrete, fibre wraps, steel belts, grouting, buttresses, external reinforcement; low cost seismic retrofitting of stone houses** (Latur, Kutch)
- **Low-cost seismic isolation** for low-rise houses (rubber pads, sliding base layers like **sand or graphite**) and **energy absorbing devices, dampers** are under research
- **New materials: bamboo and jute geo-grids, polypropylene (PP) bands** for earthen houses (**PP-band mesh in mud houses** tested by IIT Roorkee), **engineered bamboo, fibre-reinforced mortars**
- **Codes and guidelines:** BIS codes (IS 4326, 13827, 13828, 13935, 15988), **NDMA retrofitting guidelines, BMTPC Vulnerability Atlas, Guidelines for Improving Earthquake Resistance of Housing (NDMA)**
- **Awareness and training: mason training, Model house demonstration, Building Centres, Disaster Resistant Technologies in PMAY-G, Early Warning and seismic microzonation** (cities)
- **Gaps:** poor enforcement, **no mandatory approval in villages**, cost, lack of skilled people, low insurance coverage

9. Floods

Floods are caused by heavy rain, river overflow, dam breaks, drainage congestion and storm surges. Major flood zones: **Brahmaputra, Ganga, Kosi (Bihar), Mahanadi, Godavari and Krishna deltas; Assam, Bihar, UP, West Bengal, Odisha.**

Flood resistant construction

Measure	Detail
Site selection	Away from rivers, **above High Flood Level (HFL); flood plain zoning**
Raised plinth / platform	**Plinth above HFL (at least 0.6–1.0 m)** on **raised earth mounds (flood-proof platforms, villages on mounds); stilt houses on piles / bamboo posts** (Assam)
Foundation	**Deep, with scour protection; stone / brick with RCC, pile / pier foundations**; protect with **stone pitching**
Materials	**Burnt brick / stone / RCC, stabilised soil, waterproof plaster; no mud walls in flood area; bamboo, timber treated**
Services	Electrical points high up; **raised platform for toilets and hand pumps; sealed septic tanks**
Shelter	**Multi-purpose flood shelters on stilts, with a school / community**
Flood-proof houses	**Floating houses** (amphibious), **houses with openings to allow water to pass (wet-proofing)**
Drainage	Proper drains and **embankments, early warning**

10. Cyclones

Cyclones bring **strong winds (up to 200+ km/h)**, heavy rain and **storm surge** along the east coast (Andhra, Odisha, Tamil Nadu, Bengal) and Gujarat. Houses are damaged by **wind pressure and suction on the roof**, flying debris and flooding.

Wind pressure (IS 875 Part 3)

$V_z = V_b \, k_1 \, k_2 \, k_3 \, k_4$ (design wind speed; V_b = basic wind speed: 33 - 55 m/s)

$p_z = 0.6 \, V_z^2$ (N/m^2); Force = $(C_{pe} - C_{pi}) \, A \, p_z$

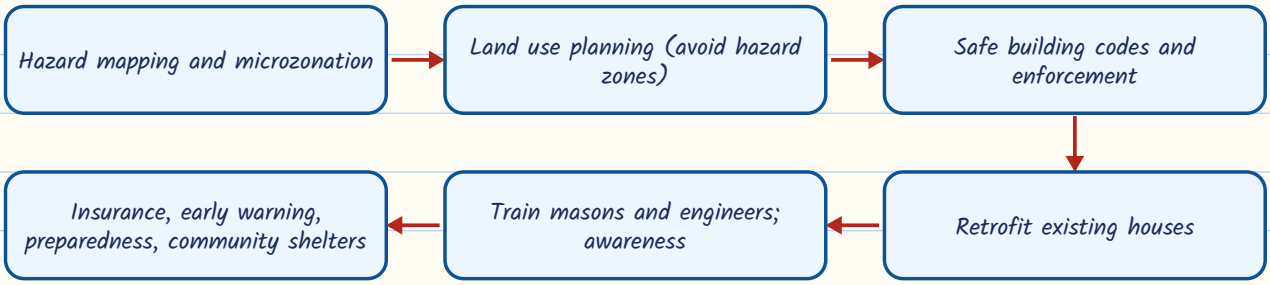
Example: V_z = 50 m/s => $p_z = 0.6 \times 2500 = 1500$ N/m^2 (1.5 kN/m^2); roof suction may be 1.0 - 1.5 times this value.

- **Cyclone resistant design: compact, square / octagonal plan, hip roof with slope 25–30 degrees** (gable ends are weak), **small eave projection (not more than 500 mm)**, low height; **strong roof-to-wall connection: tie-down bolts, straps,**

"**hold-fasts**" to the plinth beam or foundation; **strong cross bracing of roof; fix GI sheets with J-bolts and washers; avoid large openings on windward side; shutters**

- **Foundation** deep enough, **RCC columns and walls, brick / concrete walls rather than mud; plant wind-breaks (trees)** (shelter belts), **coastal embankments**
- **Cyclone shelters** (multi-purpose: raised, strong, with water, sanitation) - **about 1 per 5000 people** along coast; **early warning (IMD) and evacuation**
- **Storm surge:** houses on **raised plinth / platforms, mangrove belt**, avoid construction within the coastal regulation zone (**CRZ**)

11. Future Safety: Mitigation and Planning



- **Land use zoning and regulation**: no development in floodways, landslide slopes, coastal hazard zones; set backs
- **Compulsory use of codes (BIS, NBC 2016), techno-legal regime, third-party checks, licensed engineers and masons; disaster resistant features in all government housing (PMAY, IAY)**
- **Retrofitting** of critical buildings: schools, hospitals, shelters; **housing insurance** (e.g., **National Agricultural / rural Housing insurance**), **disaster risk transfer**
- **Community-based disaster preparedness, mock drills, training "Aapda Mitra", community participation, women involvement**
- **Use of low cost technology**: stabilised blocks, bands, bamboo, ferrocement, **light roofs**, and **owner-driven reconstruction** with technical help; **research and development, information systems (GIS, remote sensing)**

Remember for exam

* Seismic zones: II (0.10), III (0.16), IV (0.24), V (0.36); 59% of India is earthquake prone
* $V_B = A_h W$, with $A_h = (Z/2)(I/R)(S_a/g)$; heavy roof = heavy force

* Failures: diagonal cracks, corner separation, out-of-plane collapse, roof fall, gable failure
* Repair: grouting, stitching, splints and bandages, mesh jacketing; bands + vertical bars + light roof for new houses
* Precast units: bearing 75-100 mm, bond beam, screed with mesh, shear keys, good ties
* Floods: raise plinth above HFL, stilts; Cyclone: hip roof 25-30 degrees, tie-downs, shelters

Important questions

1. Explain the causes and damages to houses in an earthquake. Classify the seismic zones of India.
2. Describe the types of damages and failures of non-engineered buildings.
3. Explain the repair and restoration techniques for earthquake-damaged masonry buildings.
4. What are the recommendations for future construction in seismic areas (IS 4326)?
5. Explain the requirements of structural safety of thin precast roofing units against earthquake forces.
6. Discuss the status of R&D; in earthquake strengthening measures.
7. Describe flood and cyclone resistant features in housing.
8. Explain the future safety measures for housing in disaster prone areas.